Supervyse OPF
RAID OOB Management

Revision RV24.06

06/2024

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2023-03-25	JoeYJ Yang	Derick Lai	Initial Release

1 Introduction

This document describes how to do the customizing.
Supervyse OPF Storage System can provide the storage information on the host. The Storage System that we currently support are RAID card. In the deployment diagram(Figure 1), you can see the whole picture of the storage system. Outside the BMC, user can use the web-based storage manager and the RESTFul API based on the Redfish Storage schema to access storage information through the internet. Inside the BMC, user can develop a application that uses the Message-Interface APIs to access storage information. Message-Interface(Mi) is essentially a DBus and we implement it for portability. Whether you use inside or outside method to access data, the request messages will be pass to RAIDService. RAIDService is a key role in the Storage System. It knows how to communicate with the RAID cards by the vendor library.

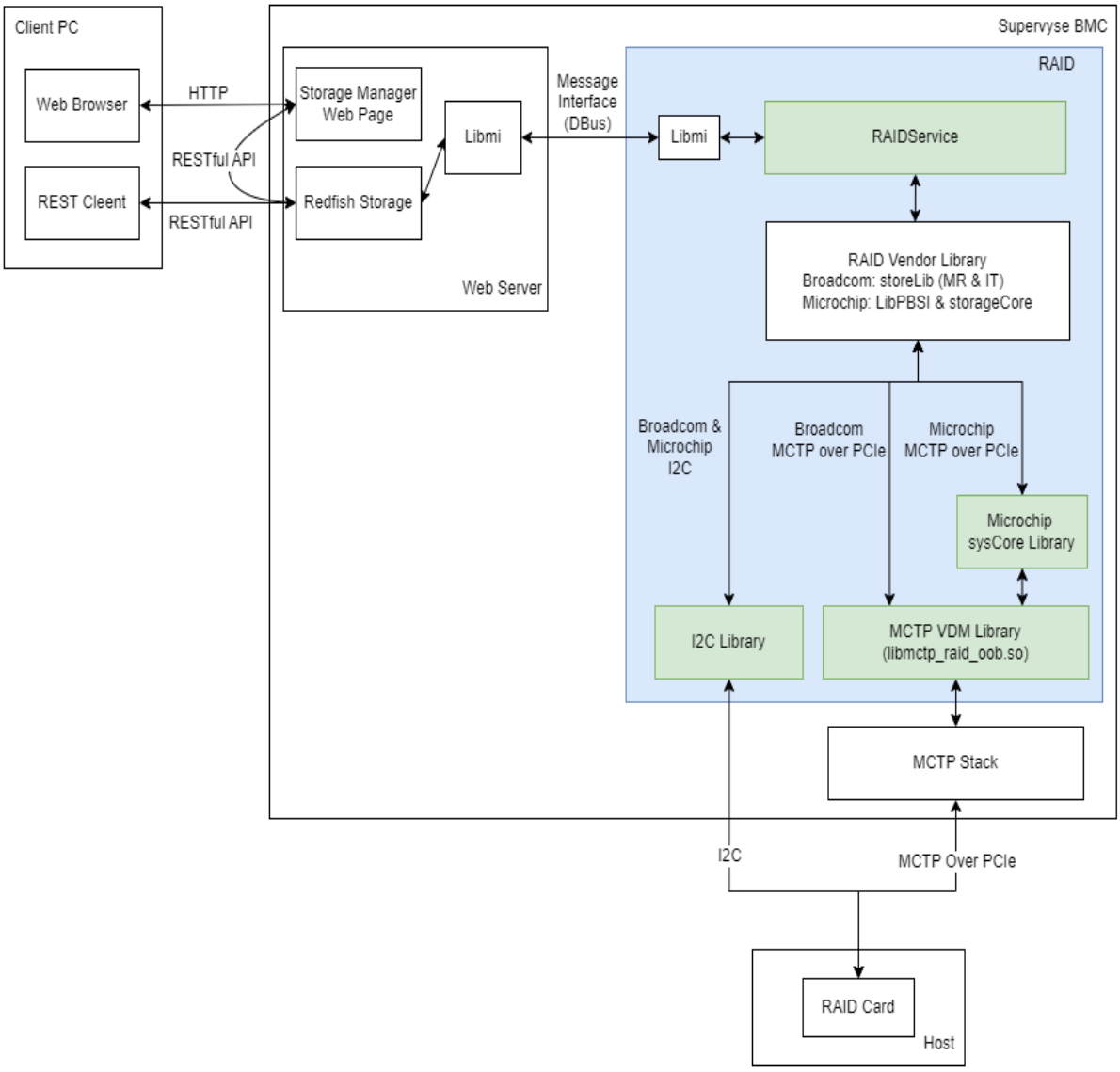


Figure 1. Deployment diagram

What is a Controller Card

A controller card is a device that sits between the host system and the storage system, and allows the two systems to communicate with each other. There are two types of controller cards: Host Bus Adapters (HBAs), and RAID controller cards.

HBA

An HBA is an expansion card that plugs into a slot (such as PCI-e) on the computer system's motherboard and provides fast, reliable non-RAID I/O between the host and the storage devices. HBAs can reliably connect hundreds or even thousands of hard disk drives (HDDs), tape, and solid-state drives (SSDs) to the host, making them ideal for cost-sensitive tape backup solutions or high-performance SSD environments.

RAID controller card

A RAID controller card is similar to an HBA, but can also add redundancy (RAID), optimize performance, reduce latency, or even make smart decisions on whether to store data on an HDD or an SSD cache, depending on user needs. Since these additional tasks consume power and processing speed, RAID controllers are typically more expensive than HBAs and handle fewer devices. Both GET/SET functions are supported for Broadcom RAID cards.

Get:
- Get Controller Information
- Get Drive Information
- Get Volume Information

Set
- Create/Delete Virtual Drive
- Scan/Import/Clear Foreign Configuration
- Global/Dedicated/Remove Hot Spare

Microchip RAID / HBA cards and Broadcom HBA card only support GET functions.

Get functions:
- Get Controller Information
- Get Drive Information

2 Configuration

There are 2 RAID OOB mode, such as **I2C mode** and **MCTP over PCIe mode**. The default RAID OOB mode is **I2C mode**. Please configure the RAID OOB mode in .config by setting **CONFIG_INSYDE_REDFISH_RAID_STORAGE_I2C_MODE** as y.

The path for your reference. ${OPF_ROOT}/meta-insyde/meta-common/meta-core/recipes-raid/raid/files/.config.

```
CONFIG_INSYDE_REDFISH_RAID_STORAGE_I2C_MODE=y
```

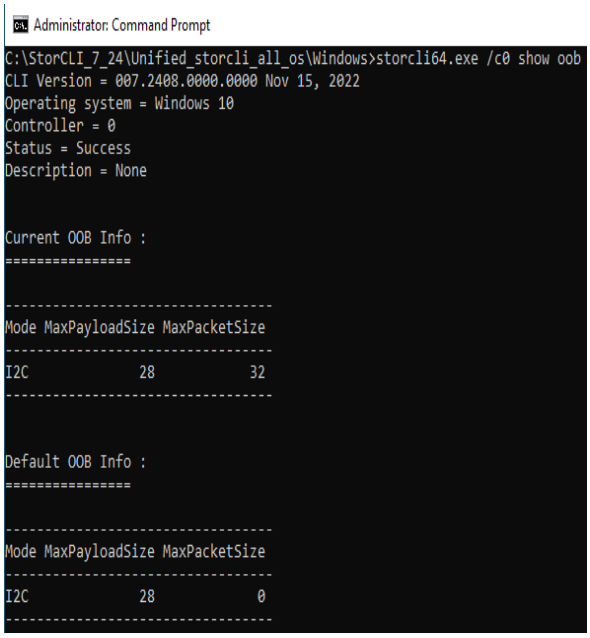
2.1 OOB mode - I2C

StorageSystem.xml

Please refer to **Chapter 4 StorageSystem.xml**

Check the firmware version of RAID cards

- **Broadcom RAID / HBA**
  - Use Broadcom's tool ***storcli*** to check firmware version, you can download it from Broadcom website. e.g. https://www.broadcom.com/products/storage/raid-controllers
  - Latest StorCLI (MR7.24)



  - Here is the command to change OOB mode as I2C in Windows Command Prompt, take Broadcom RAID 9560-16i for example.

```
Storcli64.exe /c0 set oob mode=i2c maxpayloadsize=28 maxpacketsize=32
```

- **Microchip RAID / HBA**
  - Use Microchip's tool ***arcconf*** to check the firmware of Microchip RAID card, you can download it from Microchip Web. https://storage.microsemi.com/en-us/downloads/storage_manager/sm/productid=asr-3154-16i&dn=microsemi+adaptec+smartraid+3154-16i.php
  - Adaptec ARCCONF Command Line Utility 4.09.00.25611

Select Administrator: Command Prompt

```
C:\Microchip_arcconf\cmdline>arcconf.exe GETPBSICONFIG 1
Controllers found: 1
PBSI Frequency is 400 KHZ.
PBSI Slave Address is 0xde.
PBSI Clock Stretch is Enabled.

Command completed successfully.
```

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Microsemi Adaptec SmartRAID 3154-16i

Storage Manager Downloads

Date	Filename	Description
24 Jul 2024	Adaptec bootUSB Utility 4.18.00.26842	Bootable USB image of maxView and ARCCONF Command Line Interface
24 Jul 2024	Adaptec Event Monitor 4.18.00.26842	Adaptec Event Monitor for Windows, Linux, and FreeBSD
24 Jul 2024	Adaptec ARCCONF Command Line Utility 4.18.00.26842	Adaptec CLI for Windows, Linux, Vmware, FreeBSD, and Solaris
24 Jul 2024	maxView Storage Manager 4.18.00.26842 for Microsoft Hyper-V	maxView Storage Manager for 64-bit Microsoft Hyper-V
24 Jul 2024	maxView Storage Manager 4.18.00.26842 for Citrix XenServer	maxView Storage Manager for 64-bit Citrix XenServer
24 Jul 2024	maxView Storage Manager 4.18.00.26842 for VMware	maxView Storage Manager for VMware
24 Jul 2024	maxView Storage Manager 4.18.00.26842 for Linux x64	maxView Storage Manager for 64-bit Linux
24 Jul 2024	maxView Storage Manager 4.18.00.26842 for Windows x64	maxView Storage Manager for 64-bit Windows

2.2 OOB mode - MCTP over PCIe

Microchip RAID / HBA cards do not support MCTP over PCIe.

How to set MCTP over PCIe

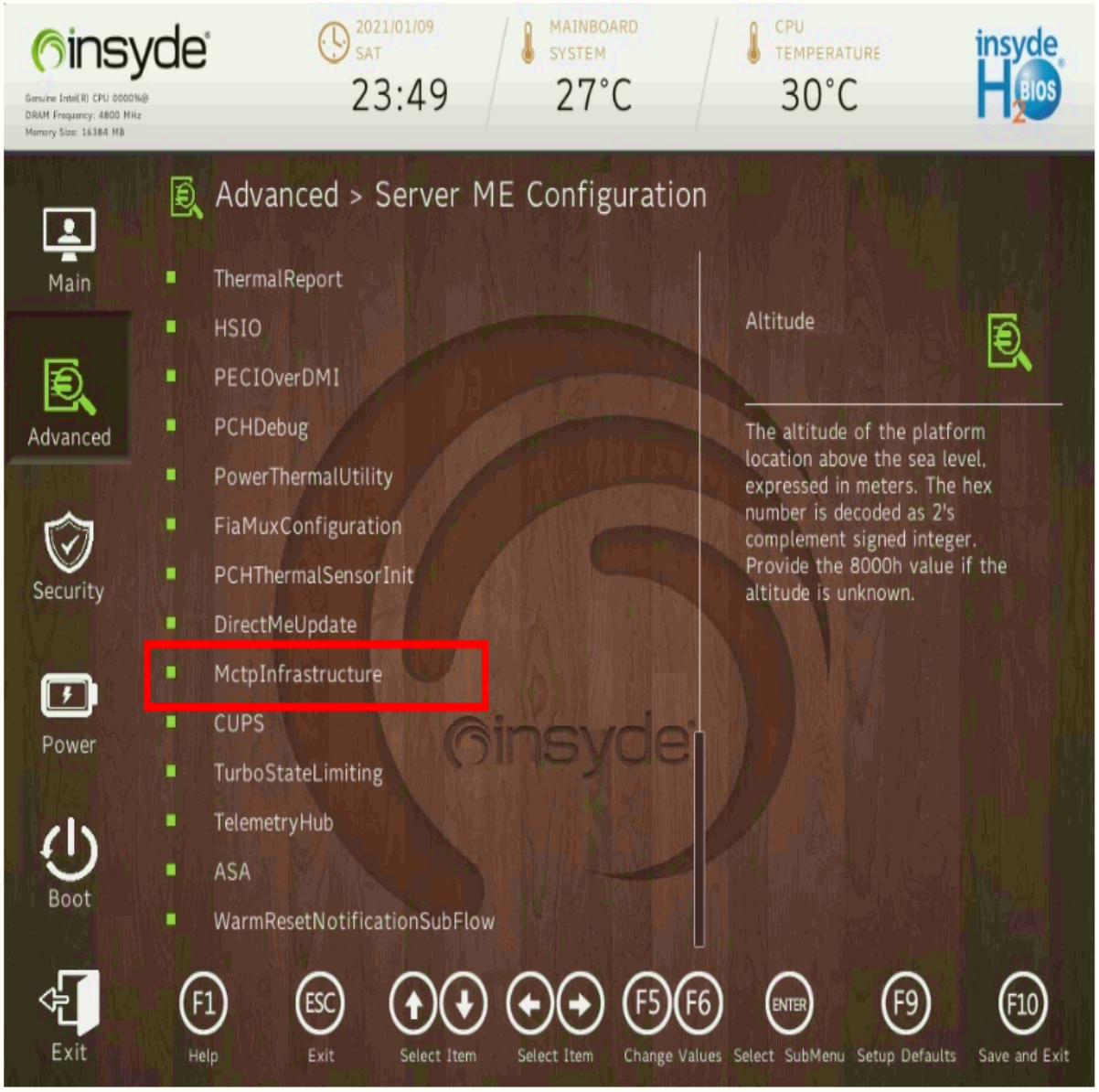
The default OOB mode is I2C. If you would like to change to MCTP over PCIe, please modify by menuconfig.
Modify following config file

```
# vi $SDK/meta-insyde/meta-common/meta-core/recipes-raid/raid/files/.config

=> CONFIG_INSYDE_REDFISH_RAID_STORAGE_MCTP_MODE=y
```

```
#
# Storage OOB Management
#
CONFIG_INSYDE_RAID_STORAGE_OOB=y
# CONFIG_INSYDE_REDFISH_RAID_STORAGE_I2C_MODE is not set
CONFIG_INSYDE_REDFISH_RAID_STORAGE_MCTP_MODE=y
```

2.2.1 Check BIOS

BIOS should support MCTP Infrastructure.



2.2.2 Check BMC

BMC should have MCTP stack.
If we can find MCTP device, BMC should support MCTP stack.

```
# busctl tree xyz.openbmc_project.MCTP_PCIe
```

```
root@intel-obmc:~# busctl tree xyz.openbmc_project.MCTP_PCIe
`-/xyz
  `-/xyz/openbmc_project
    `-/xyz/openbmc_project/mctp
      `-/xyz/openbmc_project/mctp/device
        |-/xyz/openbmc_project/mctp/device/29
        |-/xyz/openbmc_project/mctp/device/61
        |-/xyz/openbmc_project/mctp/device/8
        `-/xyz/openbmc_project/mctp/device/9
```

busctl introspect xyz.openbmc_project.MCTP_PCIe /xyz/openbmc_project/mctp/device/9

```
root@intel-obmc:~# busctl introspect   xyz.openbmc_project.MCTP_PCIe /xyz/openbmc_project/mctp/device/9
NAME                                    TYPE      SIGNATURE RESULT/VALUE                            FLAGS
org.freedesktop.DBus.Introspectable     interface -         -                                       -
.Introspect                             method    -         s                                       -
org.freedesktop.DBus.Peer               interface -         -                                       -
.GetMachineId                           method    -         s                                       -
.Ping                                   method    -         -                                       -
org.freedesktop.DBus.Properties         interface -         -                                       -
.Get                                    method    ss        v                                       -
.GetAll                                 method    s         a{sv}                                   -
.Set                                    method    ssv       -                                       -
.PropertiesChanged                      signal    sa{sv}as  -                                       -
xyz.openbmc_project.Common.UUID         interface -         -                                       -
.UUID                                   property  s         "00000000-0000-0000-0000-000000000000"  emits-change
xyz.openbmc_project.MCTP.Endpoint       interface -         -                                       -
.Mode                                   property  s         "xyz.openbmc_project.MCTP.Base.Bindin... emits-change
.NetworkId                              property  q         0                                       emits-change
xyz.openbmc_project.MCTP.PCIVendorDefined interface -       -                                       -
.MessageTypeProperty                    property  aq        1 4322                                  emits-change
.VendorID                               property  s         "0x1000"                                emits-change
xyz.openbmc_project.MCTP.SupportedMessageTypes interface -  -                                       -
.Ethernet                               property  b         false                                   emits-change
.MctpControl                            property  b         true                                    emits-change
.NCSI                                   property  b         false                                   emits-change
.NVMeMgmtMsg                            property  b         false                                   emits-change
.PLDM                                   property  b         false                                   emits-change
.SPDM                                   property  b         true                                    emits-change
.VDIANA                                 property  b         false                                   emits-change
.VDPCI                                  property  b         true                                    emits-change
```

2.2.3 Check Device ID

Find the device id of RAID card in List of PCI ID's (https://pci-ids.ucw.cz/v2.2/pci.ids) and check it in following .ini file.

1. Broadcom RAID card

Check DEVICEID in "/nv/storelibconf.ini".

```
# DeviceID for Gen-4 ctrl^M
#0x10E1, 0x10E2, 0x10E5, 0x10E6^M
DEVICEIDS=0x5B,0x5D,0x5F,0x14,0x15,0x16,0x17,0x1B,0x1C,0x10E1,0x10E2^M
```

ex. The 9560-16i device id is 0x10E2.

2. Broadcom HBA card
Check DEVICEID in "/nv/slit_bmc_conf.ini".

```
# Gen-3.5 IT Controller device IDs list
# 0xAA 0xAB 0xAC 0xAD 0xAE 0xAF 0XD1
DEVICEIDS=0xAC 0xAF 0xE5 0xE6
```

ex. The 9500-16i device id is 0xE6

2.2.4 Check RAID card Firmware

RAID card should work on OOB mode MCTP over PCIe.
1. Broadcom RAID

Use Broadcom Tool "storcli", please download the tool on Broadcom Web.
(https://www.broadcom.com/products/storage/raid-controllers)

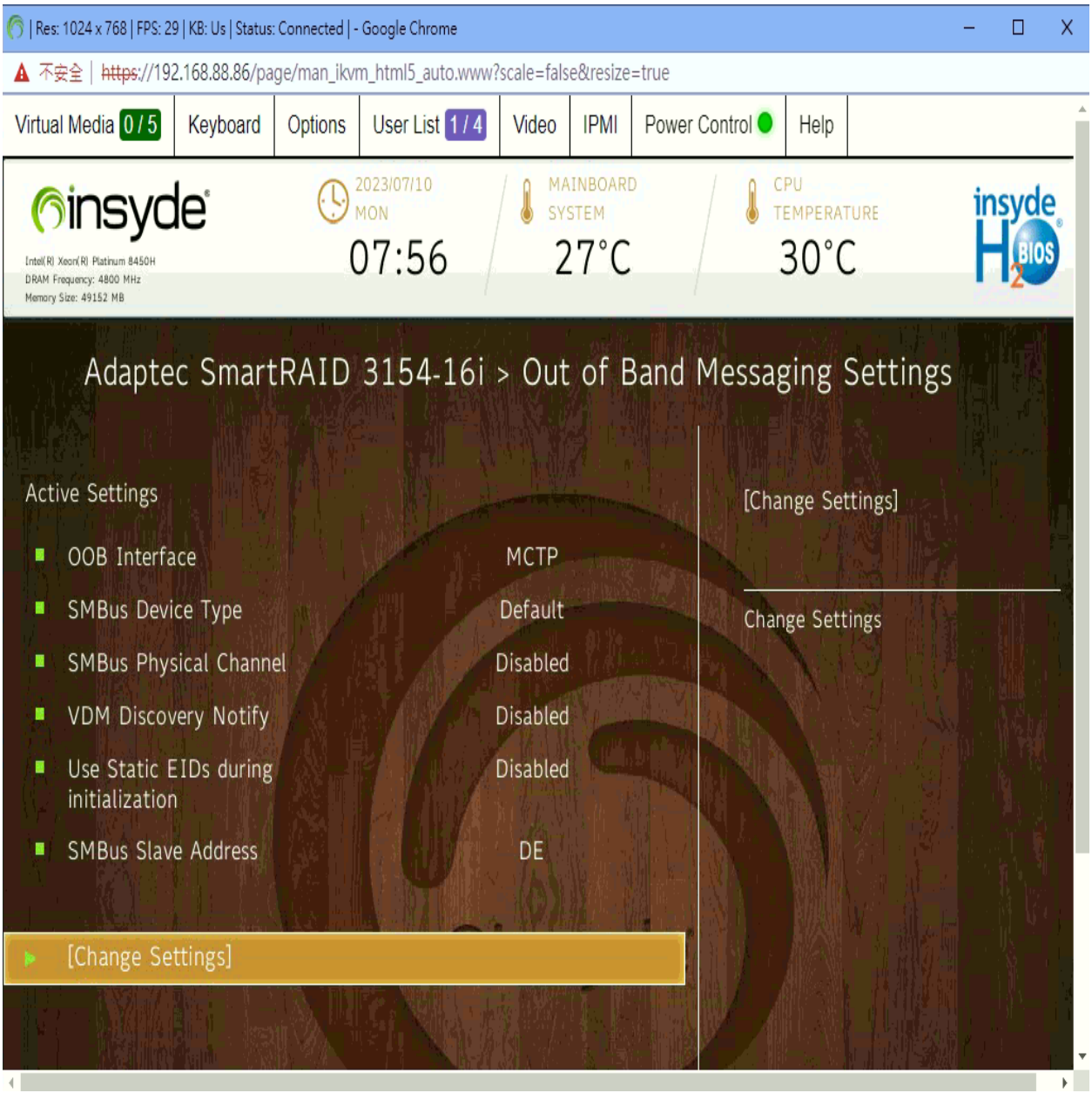
Ex. Latest StorCLI (MR7.24)

```
Select Administrator: Command Prompt
C:\StorCLI_7_24\Unified_storcli_all_os\Windows>storcli64.exe /c0 show oob
CLI Version = 007.2408.0000.0000 Nov 15, 2022
Operating system = Windows 10
Controller = 0
Status = Success
Description = None


Current OOB Info :
================

-----------------------------------
Mode MaxPayloadSize MaxPacketSize
-----------------------------------
PCIE             64            80
-----------------------------------


Default OOB Info :
================

-----------------------------------
Mode MaxPayloadSize MaxPacketSize
-----------------------------------
I2C              28             0
-----------------------------------
```

1. If current OOB mode is I2C, please modify RAID OOB mode to MCTP over PCIe by "storcli" (on Windows Command Prompt).

ex. Broadcom RAID 9560-16i
Storcli64.exe /c0 set oob mode=pcie maxpayloadsize=64 maxpacketsize=80

2. Microchip RAID
Go to BIOS menu to check or set OOB Interface "MCTP".


| Res: 1024 x 768 | FPS: 29 | KB: Us | Status: Connected | - Google Chrome
不安全 | https://192.168.88.86/page/man_ikvm_html5_auto.www?scale=false&resize=true
Virtual Media 0 / 5
Keyboard
Options
User List 1 / 4
Video
IPMI
Power Control
Help
insyde
Intel(R) Xeon(R) Platinum 8450H
DRAM Frequency: 4800 MHz
Memory Size: 49152 MB
2023/07/10
MON
07:56
MAINBOARD
SYSTEM
27°C
CPU
TEMPERATURE
30°C
insyde H2 BIOS
Adaptec SmartRAID 3154-16i > Out of Band Messaging Settings
Active Settings
OOB Interface
MCTP
SMBus Device Type
Default
SMBus Physical Channel
Disabled
VDM Discovery Notify
Disabled
Use Static EIDs during initialization
Disabled
SMBus Slave Address
DE
[Change Settings]
[Change Settings]
Change Settings

3 StorageSystem.xml

StorageSystem.xml is the configure file for overall RAID service, such as the setting for I2C channels.

In OPF SDK

${OPF_ROOT}/meta-insyde/meta-common/meta-core/recipes-raid/raid/files/StorageSystem.xml

In BMC run-time console

/nv/StorageSystem.xml.

```
<StorageSystem>
   <Storage Id="1" Name="Storage System" Description="Storage System" GeneratorId="0x41"
      PCIeI2CLogicalChannelList="16,17,18,19" DefaultI2CSlaveAddrList="0x2,0xDE,0xA" />
      <Controller Index="1" NumberOfSlots="8 StartingSensorNumber="0x5e" />
      <Controller Index="2" NumberOfSlots="8" StartingSensorNumber="0x66" />

      <Event Code="PD_INSERTED" Dir="ASSERT" Severity="Non_Critical" LED="Blink_Green" />
      <Event Code="PD_REMOVED" Dir="DEASSERT" Severity="Critical" LED="Blink_Amber" />
      <Event Code="PD_REBUILD_IN_PROG" Dir="ASSERT" Severity="Critical" LED="Blink_Amber" />
      <Event Code="PD_REBUILD_ABORT" Dir="DEASSERT" Severity="Non_Critical" LED="Blink_Green" />
      <Event Code="PD_REBUILD_DONE" Dir="ASSERT" Severity="Non_Critical" LED="Blink_Green" />
      <Event Code="LD_DEGRADED" Dir="ASSERT" Severity="Critical" LED="Blink_Amber" />
</StorageSystem>
```

3.1 The format in StorageSystem.xml.

Storage

Property	Description
Id	A unique number of the storage. The Id of the first Storage is "1".
Name	The name of the storage.
Description	The description of the storage.
PCIeI2CLogicalChannelList	The I2C logical channel.
DefaultI2CSlaveAddrList	The default I2C slave address of RAID card.

Controller

Property	Description
Index	A unique number of each RAID card. The Index of the first Controller is "1".
NumberOfSlots	The number of HDDs for each RAID card.
StartingSensorNumber	Sensor number with each HDDs.

Event

Property	Description
Code	Which event RAID card can detect and send SEL Log
Dir	Assert / De-assert
Severity	Non-Critical / Critical
LED	The behavior of System LED

In DefaultI2CSlaveAddrList and OEMI2CSlaveAddrList, the I2C slave address should not be the same.
Note: To figure out what i2c channel number is and to fill in PCIeI2CLogicalChannelList, you can use i2cdect tool to gather info. For more info, please see below.

```
root@evb-ast2600-insyde:~# i2cdetect -l
i2c-0   i2c         1e78a080.i2c-bus            I2C adapter
i2c-1   i2c         1e78a100.i2c-bus            I2C adapter
i2c-2   i2c         1e78a180.i2c-bus            I2C adapter
i2c-3   i2c         1e78a200.i2c-bus            I2C adapter
i2c-4   i2c         1e78a280.i2c-bus            I2C adapter
i2c-5   i2c         1e78a300.i2c-bus            I2C adapter
i2c-6   i2c         1e78a380.i2c-bus            I2C adapter
i2c-7   i2c         1e78a400.i2c-bus            I2C adapter
i2c-8   i2c         1e78a480.i2c-bus            I2C adapter
i2c-9   i2c         1e78a500.i2c-bus            I2C adapter
i2c-12  i2c         1e78a680.i2c-bus            I2C adapter
i2c-13  i2c         1e78a700.i2c-bus            I2C adapter
i2c-14  i2c         1e78a780.i2c-bus            I2C adapter
```

```
i2c-15  i2c       1e78a800.i2c-bus          I2C adapter
i2c-16  i2c       i2c-8-mux (chan_id 0)       I2C adapter <--- This is i2c-8 Mux channel 0
i2c-17  i2c       i2c-8-mux (chan_id 1)       I2C adapter <--- This is i2c-8 Mux channel 1
i2c-18  i2c       i2c-8-mux (chan_id 2)       I2C adapter <--- This is i2c-8 Mux channel 2
i2c-19  i2c       i2c-8-mux (chan_id 3)       I2C adapter <--- This is i2c-8 Mux channel 3 that raid card use here
```

4 Verification

4.1 Microchip

MicrosemiTool

This is a tool provided by Microchip which run on BMC console, and it provides much more debug information.

```
root@evb-ast2600-insyde:~# MicrosemiTool
Version 1.0.0

Usage: (Please use 0x in the parameters.)
   MicrosemiTool <I2CBus> <DeviceSlaveAddr> <MuxAddr> <MuxData> <2nd MuxAddr> <2nd MuxData> ...

Ex. If you have I2C Mux
   MicrosemiTool 0x2 0xde 0xe0 0x4

Ex. If you don't have I2C Mux
   MicrosemiTool 0x2 0xde 0x0 0x0
```

```
root@evb-ast2600-insyde:~# MicrosemiTool 0x13 0xde 0x00 0x00
I2C Bus        : 0x13
I2C Slave Addr : 0xde
I2C Mux AddrBus: 0x0
I2C Mux Data   : 0x0
I2C Mux Level  : 0x1

PBSI Version:
========================================
pbsi major version: 01
pbsi minor version: 05
dst major vesrion: 01
dst minor vesrion: 05
========================================

RAID Card Vital Stats:
========================================
pbsi status = 0x1 (I2C Ready)
adapter status = 0x2 (DST initialization complete)
adapter status change = 0x0
ambient temperature = 33 degrees C
junction temperature = 44 degrees C
========================================

RAID Card Adapter Info:
========================================
type = 0x0012
name = 3154-16i
serial number = 722963D6198
flags = 2
loaded basecode version = 06.05.008.00
Firmware version = 3.21[0]
Staged Firmware version = 3.21[0]
CPLD (Load version/Flash version) = 5/0
SEEPROM (Load version/Flash version) = 2/0
```

```
PCI Vendor ID = 0x9005
PCI Device ID = 0x1380
PCI Subvendor ID = 0x028F
PCI Subsystem ID = 0x9005
DRAM size = 4096 MB
WWN = 50000D170384D900
Ports (Internal/External) = 4/0
========================================

Green Backup Info:
========================================
GB status = 0x600
      learning cycle failed
      super capacitor pack failed
max temperature = 60 degrees C
max voltage = 4600 V
current temperature = 22 degrees C
current voltage = 4600 mV
active current draw = 980 mA
serial number =
temperature threshold = 50 degrees C
health = 100%
charge state = 0%
========================================

RAID Card Topology Info:
========================================
number of logical devices = 0
number of failed or offline logical devices = 0
number of physical devices = 3
number of failed or offline physical devices = 0
========================================

Physical devices information available at index 0
========================================
vendor = ATA
product id = WDC WD5000BPKT-2
revision level = 01.0
revision level sata = 1A01
serial number = WD-WXK1E72DPDT4
device usage = 16
device state = 1
device error_count = 1
device capacity = 976773168 blocks
device block Size = 512 bytes
device Size = 465 GB
temperature = 35 degrees C
type = SATA HDD (3 Gbps)
========================================

Physical devices information available at index 1
========================================
vendor = ATA
product id = WDC WD5000BPKT-2
revision level = 01.0
revision level sata = 1A01
serial number = WD-WXD1E91NXKV9
device usage = 16
device state = 1
device error_count = 30
device capacity = 976773168 blocks
```

```
device block Size = 512 bytes
device Size = 465 GB
temperature = 36 degrees C
type = SATA HDD (3 Gbps)
=========================================

Physical devices information available at index 2
=========================================
vendor = ATA
product id = TOSHIBA MK5059GS
revision level = GT00
revision level sata = 1H
serial number = X1PHP0Y6T
device usage = 16
device state = 1
device error_count = 1
device capacity = 976773168 blocks
device block Size = 512 bytes
device Size = 465 GB
temperature = 34 degrees C
type = SATA HDD (3 Gbps)
=========================================
```

2. NG situation

```
root@evb-ast2600-insyde:~# MicrosemiTool 0x13 0xde 0x00 0x00
I2C Bus        : 0x13
I2C Slave Addr : 0xde
I2C Mux AddrBus: 0x0
I2C Mux Data   : 0x0
I2C Mux Level  : 0x1
GetPBSIVersion failed!
GetVitalStats failed!
GetAdapterInfo failed!
GetGreenBackupInfo failed!
GetLDTaskProgressInfo failed!
GetTopologyInfo failed!
```

4.2 Broadcom

In BMC side, there is a tool named "StoreLibTest" you could use.

3. The HELP of StoreLibTest

```
root@evb-ast2600-insyde:~# StoreLibTest
Usage: (Please use 0x in the parameters.)
   StoreLibTest <I2CBus> <DeviceSlaveAddr> <MuxAddr> <MuxData> <2nd MuxAddr> <2nd MuxData> ...

Ex. If you have I2C Mux
   StoreLibTest 0x2 0x2 0xe0 0x2

Ex. If you don't have I2C Mux
   StoreLibTest 0x2 0x2 0x0 0x0
```

4. Good situation

```
root@evb-ast2600-insyde:~# StoreLibTest 0x13 0x02 0x00 0x00
Library loaded for MegaRAID SAS Controller.
Failed to load library for SAS IT/IR Controller.
dlerror = libstorelibir.so: cannot open shared object file: No such file or directory
Failed to load library for SAS IT/IR2 Controller.
dlerror = libstorelibir-2.so: cannot open shared object file: No such file or directory
Failed to load library for SAS IT/IR3 Controller.
```

```
dlerror = libstorelibir-3.so: cannot open shared object file: No such file or directory
Library loaded for SAS IT Controller.
Storelibtest version 2.78

Storelib Version : 07.1701.0100.0000

StorelibIT Version : 07.1800.0200.0000


Initializing library for MegaRAID SAS Controller...
argc=5

CtrlId[0]=0x1302
ChannelAddress[0]=0x13  DeviceAddress[0]=0x2
I2CMuxAddr[0]=0x0       I2CMuxData[0]=0x0


Do you want to set OOB transport type ?  (Type 'y' for yes, 'n' for no)--->y
Enter OOB Transport Type[Default=0/PCIe=1/I2C=2]:
2
SetOOBTransportType Success
InitializeLibraryMR successful

MegaRAID SAS Controller is running under OOB Mode.



Initializing library for SAS IT Controller...
argc=5

CtrlId[0]=0x213
ChannelAddress[0]=0x13  DeviceAddress[0]=0x2
I2CMuxAddr[0]=0x0       I2CMuxData[0]=0x0


Do you want to set OOB transport type ?  (Type 'y' for yes, 'n' for no)--->y
Enter OOB Transport Type[Default=0/PCIe=1/I2C=2]:
2
SetOOBTransportType Success
InitializeLibraryIT successful


UnInitializing library for SAS IT Controller...
UnInitializeLibrary successful
# of controllers found = 1
Index       Controller ID       Controller Type           Product name            SAS Address
-----       -------------       ---------------           ------------            -----------
  0          4866          MegaRAID SAS ControllerAVAGO MegaRAID SAS 9460-8i      500062b203f8aa00
Only one controller present. Selected Controller = 0


         ...MAIN MENU...
         ---------------
(1) SelectController    (2) System
(3) Controller          (4) Physical Device
(5) Logical Device      (6) Configuration
(7) BBU                 (8) Passthru
(9) Event               (10)Enclosure
(11)Run BVT             (12)Security
(13) OOB                (14) StressTest
```

```
0 TO QUIT
Please enter choice : 3
                ...CONTROLLER MENU...
                ----------------------
(1)  GetControllerInfo            (2)  GetControllerProperties
(3)  SetControllerProperties      (4)  GetPhysicalDevicesList
(5)  GetEnclosures List           (6)  GetTopologyInfo
(7)  SLT_GetAlarm                 (8)  SLT_EnableAlarm
(9)  SLT_DisableAlarm             (10) SLT_SilenceAlarm
(11) FlashControllerFirmware      (12) SLT_GetPCIInfo
(13) GetLogicalDriveDistribution  (14) GetLogicalDrivesList
(15) SLT_GetPatrolReadStatusInfo      (16) SLT_GetPatrolReadProperties
(17) SLT_SetPatrolReadProperties      (18) SLT_StartPatrolReadOperation
(19) SLT_StopPatrolReadOperation      (20) SLT_GetDriverVersion
(21) SetRebuildRate               (22) GetControllerHealth
(23) GetPhysicalDriveDistribution (24) StartSelfCheckDiagnostics
(25) SLT_GetAllLDsAllowedOperation    (26) GetMFCDefaults
(27) GetSasConnectorInfo          (28) SetSasConnector
(29) DisplayPhyErrorInfo          (30) GetCCScheduleStatus
(31) GetCCScheduleProperties      (32) SetCCScheduleProperties
(33) GetMFCManufacturingInfo      (34) GetMFCManufacturingInfo2
(35) FlashPDFirmware              (36) FlashPDFirmwareSTP
(37) GetPinnedCacheList           (38) DiscardPinnedCache
(39) ClearBBMLog                  (40) GetPDQueryList
(41) ShowAdvanceSoftwareOptions   (42) ActivateAdvanceSoftwareOptions
(43) GetSafeID                    (44) DeactivateTrialAdvanceSoftwareOptions
(45) TransferAdvanceSoftwareOptionsToKeyVault
(48) GetSerialNo                  (49) GetRehostInfo
(50) Acknowledge ReHost
(51) Schedule Power Policy
(52) GetSASPhyInfo                (53) UpdateSASPhyLinkSpeed
(54) SuspendPatrolReadOperation   (55) ResumePatrolReadOperation
(56) StartLocateZone              (57) StopLocateZone
(58) SLT_ResetController          (59) Personality Mode
(60) Set Personality Mode         (61) AutoConfigurePersonality
(62) FlashCtrlDualFirmware        (63) GetProfileList
(64) SetProfileId                 (65) SetBehaviorMode
(66) Snapdump Property            (67) Snapdump Show
(68) Snapdump Retrieve            (69) Clear snapdump
(70) TriggersOnDemandSnapDump     (71) GetOOBMode
(72) SetOOBMode                   (73) GetTTYLog
(74) GetPCIeLaneInfo              (75) GetPDFWDownloadStatus
(76) ShowAllSecurityProtocol      (77) ShowSlotStatus
(78) InvalidateCertificateSlot    (79) SLT_ExportCSR
(80) SLT_ImportCertificateChain   (81) ReadCertificate
(82) FlashEnclosureFirmware

0 TO QUIT. Invalid option to return to Previous Menu
Currently selected controller = 0
Please enter choice : 1
Controller Information
----------------------
PCI Information
        Vendor ID          : 0x1000
        Device ID          : 0x16
        SubVendor ID       : 0x1000
        SubDevice ID       : 0x9461

Host Interface Information
        Port Count         : 0
        Host Interface     : PCIE
```

```
Device Interface Information
    Port Count          : 8
    Device Interface    : SAS_3G SATA_3GSAS_6GSAS_12G
    Backend SAS addresses
        0 : 300062b203f8aa01
        1 : 300062b203f8aa03
        2 : 300062b203f8aa05

List of components residing in flash
    Image Checkword         : 0x36313533
    Image Component Count   : 7
    ImageComponent [ 0 ] :
        Name        : SBR
        Version     : 5.1800.00-0472
        Build Date  : Apr 05 2021
        Build Time  : 20:23:46
    ImageComponent [ 1 ] :
        Name        : BTBL
        Version     : 7.02.00.00-0021
        Build Date  : Jun 28 2018
        Build Time  : 16:01:28
    ImageComponent [ 2 ] :
        Name        : BIOS
        Version     : 7.18.00.0_0x07120200
        Build Date  : 05/20/2021

        Build Time  : 15:38:04
    ImageComponent [ 3 ] :
        Name        : HIIM
        Version     : 07.18.03.00
        Build Date  : May 20 2021
        Build Time  : 15:07:17
    ImageComponent [ 4 ] :
        Name        : HIIA
        Version     : 07.18.03.00
        Build Date  : May 20 2021
        Build Time  : 15:12:24
    ImageComponent [ 5 ] :
        Name        : APP
        Version     : 5.180.00-3512
        Build Date  : May 04 2021
        Build Time  : 18:05:13
    ImageComponent [ 6 ] :
        Name        : NVDT
        Version     : 5.1800.00-0001
        Build Date  : Mar 19 2021
        Build Time  : 19:30:09

Pending Image Component Count : 0

Maximum Arms per LD     : 32
Maximum Spans per LD    : 8
Maximum arrays          : 240
Maximum number of LDs   : 64
Product Name            : AVAGO MegaRAID SAS 9460-8i
Serial #                : SP82521910

Other hardware present
    Alarm
    NVRAM
```

```
        UART
        Memory
        Flash
        ROCtemperature : 73 Cel

Current Firmware Time : 3/21/2023 ; 9:23:9

Maximum Data Transfer Rates
        Maximum parallel commands supported              : 4077
        Maximum Scatter Gather Element(SGE) count allowed  : 60
        Maximum data transfer size in sectors            : 2048

Logical and Physical Device counts
        # logical devices present          : 1
        # logical devices critical         : 0
        # logical devices offline          : 0
        # physical devices present         : 4
        # physical disk devices present     : 3
        # physical disk devices critical    : 0
        # physical disk devices failed      : 0

Memory Size Information
        NVRAM size (in KB)   : 128
        Memory size (in MB)  : 2048
        Flash size (in MB)   : 16

Error Counters
        Memory correctable error count      : 0
        Memory uncorrectable error count    : 0

ShieldState Information
        Shieldstate enabled on the card

Cluster Information:
        Clustering not enabled on the card
        Cluster inactive
        Topology Type: None
        HA Multipathing : Not Supported

Supported RAID levels
        RAID level 0
        RAID level 1
        RAID level 5
        RAID level 6
        RAID level 1 with spanning
        RAID level 5 with spanning
        RAID level 6 with spanning
        RAID level 0 with spanning
        RAID level PRL 11
        RAID level PRL 11 with spanning
        SRL 3 supported

Supported Adapter options
        Rebuild Rate
        Check Consistency Rate
        BGI Rate
        Reconstruction Rate
        Patrol Rate
        Alarm Control
        Battery API enabled
        Spanning allowed
```

```
        Dedicated Hot Spares
        Revertible Hot Spares
        Foreign Config Import allowed
        Controller supports self-diagnostic
        Controller supports 'Abort on Detecting Inconsistency'
        Controller supports Multipath
        Controller supports only even drive count in RAID 1E
        Security is not supported
        FW supports personality change between the supported modes

Supported Logical Drive Operations
        Read Policy
        Write Policy
        Access Policy
        Disk Cache Policy
        Reconstruction
        LDBBM is supported

Supported Strip Sizes
        Minimum    : 64K (DDF:7)
        Maximum    : 1024K (DDF:11)

Supported Physical Drive Operations
        Force Online
        Force Offline
        Force Rebuild
        Power state : Controller will take care of spinning the drives Please Avoid spinning the drives prior to Add VD
Supported physical drive mix on controllers, encl, LD
        Controller supports SAS
        Controller supports SATA
        Allow Mix in enclosure
        Allow Mix in LD

ECC Bucket Count     : 0

Controller is not assigned a lock key


Got lock key on bootup


Lock key has been backed up


Bios was detected during boot


SSC is not supported


Controller cache is not pinned
FW Package Version   : 51.18.0-4054

Minimum PDs for RAID Level 0     : 1

Maximum PDs for RAID Level 0     : 32

Minimum PDs for RAID Level 1     : 2

Maximum PDs for RAID Level 1     : 2
```

```
Minimum PDs for RAID Level 5     : 3

Maximum PDs for RAID Level 5     : 32

Minimum PDs for RAID Level PRL 11 : 4

Maximum PDs for RAID Level PRL 11 : 32

Minimum PDs for RAID Level PRL 11 with spanning : 8

Maximum PDs for RAID Level PRL 11 with spanning : 64

Minimum PDs for RAID Level 6     : 3

Maximum PDs for RAID Level 6     : 32

Minimum PDs for RAID Level 10    : 4

Maximum PDs for RAID Level 10    : 16

Minimum PDs for RAID Level 50    : 6

Maximum PDs for RAID Level 50    : 64

Minimum PDs for RAID Level 60    : 6

Maximum PDs for RAID Level 60    : 64

Maximum supported PDs            : 64

Maximum supported dedicated HSPs : 62

Maximum supported global HSPs    : 64

Maximum supported LDs per array  : 16

DDF size <in MB>                 : 512

Partitions in DDF                : 16

FW currently does not have lock key with it.

FW supports OCE with no drive addition.

Controller Properties
---------------------
Sequence #                                : 25
# of seconds between predictive fail polls    : 300
Interrupt throttle completion count          : 16
Interrupt throttle completion time           : 50
Rebuild Rate                              : 30
Patrol Read Rate                          : 30
Back Ground Initialization(BGI) Rate         : 30
Consistency Check(CC) Rate                : 30
Reconstruction Rate                       : 30
Cache Flush Interval (in seconds)            : 4
Maximum # of drives to spin up at one time    : 2
# of seconds to delay among spinup groups     : 2
Cluster Enable                            : 0
JBOD Enable                               : 1
Spin Down Unconfigured Drive enabled         : 1
```

```
Drive capacity coercion mode                : None
Alarm Enable                                : DISABLE
Disable AutoRebuild                         : FALSE
Disable Battery Warning                     : FALSE
ECC Bucket Size                             : 15
ECC Bucket Leak Rate (in minutes)           : 1440
Restore Hotspare on Insertion               : NO
Expose Enclosure Devices                    : YES
Maintain PdFail History across reboots      : YES
Disallow Host Request Reordering            : NO
Abort CC on Detecting Inconsistency         : YES
Copyback Disabled                           : NO
Smart Copyback Enabled                      : YES
PR CorrectUnconfiguredAreas Enabled         : YES
Only allow FDE drive based encryption       : YES
Disable Ncq                                 : 0
Copyback to SSD on SMART error              : 1
SSD Patrol Read                             : 1
Use Global Spares For Emergency             : YES
Use UnconfGood For Emergency                : YES
Use Emergency Spares for SMARTer            : NO
Load Balance Mode                           : Auto load balance
Spinup Drive count                          : 2
Spinup Delay                                : 2
Disable spindown Hotspare                   : 0
Spin down time                              : 30
OFU<online FW update> is enabled
SES device VPD association type based on    : LUN
supportSnapDump                             : Enabled
Firmware Device Order                       : Disabled
Support NVMe Init failure report/drive repair : NO

Cachecade Properties
-----------------------
Cachecade - SSD Caching                     : DISABLE
Write Cache Capable                         : NO
Current Cache Size                          : 0 GB
Maximum Cache Size                          : 0 GB
```

5. NG situation

```
root@evb-ast2600-insyde:~# StoreLibTest 0x13 0x02 0x00 0x00
Library loaded for MegaRAID SAS Controller.
Failed to load library for SAS IT/IR Controller.
dlerror = libstorelibir.so: cannot open shared object file: No such file or directory
Failed to load library for SAS IT/IR2 Controller.
dlerror = libstorelibir-2.so: cannot open shared object file: No such file or directory
Failed to load library for SAS IT/IR3 Controller.
dlerror = libstorelibir-3.so: cannot open shared object file: No such file or directory
Library loaded for SAS IT Controller.
Storelibtest version 2.78

Storelib Version : 07.1701.0100.0000

StorelibIT Version : 07.1800.0200.0000



Initializing library for MegaRAID SAS Controller...
argc=5

CtrlId[0]=0x1302
ChannelAddress[0]=0x13  DeviceAddress[0]=0x2
```

```
I2CMuxAddr[0]=0x0       I2CMuxData[0]=0x0


Do you want to set OOB transport type ?  (Type 'y' for yes, 'n' for no)--->y
Enter OOB Transport Type[Default=0/PCIe=1/I2C=2]:
2
SetOOBTransportType Success
InitializeLibraryMR successful


Initializing library for SAS IT Controller...
argc=5

CtrlId[0]=0x213
ChannelAddress[0]=0x13  DeviceAddress[0]=0x2
I2CMuxAddr[0]=0x0       I2CMuxData[0]=0x0


Do you want to set OOB transport type ?  (Type 'y' for yes, 'n' for no)--->y
Enter OOB Transport Type[Default=0/PCIe=1/I2C=2]:
2
SetOOBTransportType Success
InitializeLibraryIT failed; rval = 0x8048
InitLib failed for SAS IT Controller!! rval 0x8048


UnInitializing library for SAS IT Controller...
UnInitializeLibrary failed; rval = 0x801A
ExitLib failed for SAS IT Controller!! rval 0x801A
# of controllers found = 1
Index       Controller ID       Controller Type           Product name            SAS Address
-----       -------------       ---------------           ------------            -----------
  0          4866          MegaRAID SAS ControllerFailed! rval: 0x800aOnly one controller present. Selected Controller = 0    <---
this error
```

4.3 Debug Logs

Journal Log

Please issue the command following command to dump journal logs.

```
journalctl -u raid_service
```

Here is an example that a RAID card is found.

```
root@evb-ast2600-insyde:~# journalctl -u raid.service
-- Journal begins at Fri 2023-03-24 04:02:10 UTC, ends at Fri 2023-03-24 04:07:52 UTC. --
Mar 24 04:02:14 evb-ast2600-insyde systemd[1]: Starting Supervyse raid service...
Mar 24 04:02:16 evb-ast2600-insyde systemd[1]: Started Supervyse raid service.
Mar 24 04:02:19 evb-ast2600-insyde RAIDService[250]: RAIDService version is 2.16
Mar 24 04:02:19 evb-ast2600-insyde RAIDService[250]: System Id is 123456789abcdef
Mar 24 04:02:19 evb-ast2600-insyde RAIDService[250]: libmctp_broadcom.so: cannot open shared object file: No such file or
directory
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]:      0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
```

```
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 00: 00 01 -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:28 evb-ast2600-insyde RAIDService[250]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 04:02:21 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0x2 in scan list
Mar 24 04:02:21 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:21 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xde in scan list
Mar 24 04:02:21 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:21 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xa in scan list
Mar 24 04:02:21 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0x2 in scan list
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xde in scan list
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xa in scan list
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0x2 in scan list
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xde in scan list
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xa in scan list
Mar 24 04:02:22 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:24 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0x2 in scan list
Mar 24 04:02:24 evb-ast2600-insyde RAIDService[250]: Found 0x02 in scan list          <------- A RAID Card is found.
Mar 24 04:02:24 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xde in scan list
Mar 24 04:02:24 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:24 evb-ast2600-insyde RAIDService[250]: Finding Slave Addr 0xa in scan list
Mar 24 04:02:24 evb-ast2600-insyde RAIDService[250]: Not Found
Mar 24 04:02:30 evb-ast2600-insyde RAIDService[250]: [CtrlId 4866 Successful detection]
```

6. Below is an example that no RAID cards are found.

```
root@evb-ast2600-insyde:~# journalctl -u raid.service
-- Journal begins at Fri 2023-03-24 06:03:25 UTC, ends at Fri 2023-03-24 06:03:45 UTC. --
Mar 24 06:03:29 evb-ast2600-insyde systemd[1]: Starting Supervyse raid service...
Mar 24 06:03:30 evb-ast2600-insyde systemd[1]: Started Supervyse raid service.
Mar 24 06:03:33 evb-ast2600-insyde RAIDService[246]: RAIDService version is 2.16
Mar 24 06:03:33 evb-ast2600-insyde RAIDService[246]: System Id is 123456789abcdef
Mar 24 06:03:33 evb-ast2600-insyde RAIDService[246]: libmctp_broadcom.so: cannot open shared object file: No such file or
directory
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 70: UU -- 72 -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]:     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 00: 00 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 10: 10 -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:41 evb-ast2600-insyde RAIDService[246]: 60: -- 61 -- -- 64 -- -- -- -- -- -- -- -- -- -- --
Mar 24 06:03:34 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0x2 in scan list
Mar 24 06:03:34 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:34 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xde in scan list
Mar 24 06:03:34 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:34 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xa in scan list
Mar 24 06:03:34 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0x2 in scan list
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xde in scan list
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xa in scan list
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0x2 in scan list
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xde in scan list
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xa in scan list
Mar 24 06:03:35 evb-ast2600-insyde RAIDService[246]: Not Found
```

```
Mar 24 06:03:37 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0x2 in scan list
Mar 24 06:03:37 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:37 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xde in scan list
Mar 24 06:03:37 evb-ast2600-insyde RAIDService[246]: Not Found
Mar 24 06:03:37 evb-ast2600-insyde RAIDService[246]: Finding Slave Addr 0xa in scan list
Mar 24 06:03:37 evb-ast2600-insyde RAIDService[246]: Not Found
```

RAID log on console

You can see more log on BMC console directly with below command:

```
root@evb-ast2600-insyde:~# systemctl stop raid.service
root@evb-ast2600-insyde:~# ps | grep RAID
root@evb-ast2600-insyde:~# RAIDService -c &
```

7. Example

```
root@evb-ast2600-insyde:~# dbus-send --system --print-reply --reply-timeout=60000 --dest=com.insyde.storage
/com/insyde/storage/Systems/123456789abcdef/Storage/1/StorageControllers/1
com.insyde.storage.StorageController.GetFreeRaidResource
RAID DEBUG StorageControllersGetFreeRaidResource +
RAID DEBUG StorageControllersGetFreeRaidResource System Id: 123456789abcdef, Storage Id: 1, StorageController Id: 1
RAID DEBUG UpdateController +
RAID DEBUG UpdateController mIsInit: 1
RAID DEBUG UpdateEventSummary +
RAID DEBUG UpdateEventSummary mIsInit: 1
RAID DEBUG UpdateEventSummary -
RAID DEBUG UpdateController Data is changed, update it.
RAID DEBUG UpdateControllerValue +
RAID DEBUG UpdateControllerValue -
RAID DEBUG UpdateController -
RAID DEBUG PrintConfigSummary
...All Physical Drives Information...
RAID DEBUG PrintConfigSummary Index        DeviceID      Size(MB)      State
RAID DEBUG PrintConfigSummary -----        --------      --------      -----
RAID DEBUG PrintConfigSummary 0            1             476416        UNCONFIGURED_GOOD
RAID DEBUG PrintConfigSummary 1            2             476416        UNCONFIGURED_GOOD
RAID DEBUG PrintConfigSummary 2            3             476416        UNCONFIGURED_GOOD
RAID DEBUG PrintConfigSummary

...Array Information...
RAID DEBUG PrintConfigSummary No arrays detected
RAID DEBUG PrintConfigSummary No LDs detected
RAID DEBUG StorageControllersGetFreeRaidResource -
```

4.4 Log on Library side

4.4.1 Broadcom Library

4.4.1.1 Ini file path

8. Broadcom RAID (MR)

```
/nv/storelibconf.ini
```

9. Broadcom HBA (IT)

```
/nv/slit_bmc_conf.ini
```

10. Set DEBUGLEVEL and DISABLELOG as below, then you can check log file on /var/log

```
# Debug Level(0-No Debug, 1-Level 1,2-Level 2)^M
DEBUGLEVEL=2
```

```
# Set it to disable debug logging^M
DISABLELOG=0
```

4.4.1.2 Log file path

11. Broadcom RAID (MR)

```
/var/log/storelibdebug.txt
```

12. Broadcom HAB (IT)

```
/var/log/storelibdebugit.txt
```

13. Good situation

```
Sun Nov  6 14:19:22.36 2022 : 1995943952: SendOobPacket: OOB MUTEX released
Sun Nov  6 14:19:22.36 2022 : 1995943952: SendOobPacket: Exit retVal=0x0
Sun Nov  6 14:19:22.37 2022 : 1995943952: SendOobDCMD: Exit dcmdOpCode=0x1020100 or controlCmd=0x0, retVal=0x0
Sun Nov  6 14:19:22.37 2022 : 1995943952: ProcessLibCommand: Exit: cmdType = 0x1, cmd = 0x1, retVal = 0x0
Sun Nov  6 14:19:22.38 2022 : 1995943952: ProcessLibCommand: Entry cmdType = 0x1, cmd = 0x2
Sun Nov  6 14:19:22.38 2022 : 1995943952: CSLSystem_GetCtrl: CSLSystem mutex acquired
Sun Nov  6 14:19:22.39 2022 : 1995943952: CSLSystem_GetCtrl: CSLSystem mutex released
Sun Nov  6 14:19:22.43 2022 : 1995943952: SL_ValidateInput: Controller Found [ctrlId] 0xa02
Sun Nov  6 14:19:22.44 2022 : 1995943952: CSLSystem_GetCtrl: CSLSystem mutex acquired
Sun Nov  6 14:19:22.44 2022 : 1995943952: CSLSystem_GetCtrl: CSLSystem mutex released
Sun Nov  6 14:19:22.45 2022 : 1995943952: ProcessLibCommand: Trying to acquire ctrl 0xa02 mutex
Sun Nov  6 14:19:22.45 2022 : 1995943952: ProcessLibCommand: ctrl 0xa02 mutex acquired
Sun Nov  6 14:19:22.46 2022 : 1995943952: SendDCMD: Entry ctrlId 0xa02, dcmd->opcode 0x1020200
Sun Nov  6 14:19:22.46 2022 : 1995943952: CSLSystem_GetCtrl: CSLSystem mutex acquired
Sun Nov  6 14:19:22.47 2022 : 1995943952: CSLSystem_GetCtrl: CSLSystem mutex released
Sun Nov  6 14:19:22.47 2022 : 1995943952: Fill_mbox_for_ext_support: Exit retVal 0x0
Sun Nov  6 14:19:22.47 2022 : 1995943952: SendOobDCMD: Entry ctrlId 0xa02 pDcmd->opCode= 0x1020200, pDcmd-
>dataTransferLength = 0x40, pDcmd->flags = 1
```

14. NG situation

```
Sun Nov  6 13:23:38.394 2022 : 1995837456: SendOobPacket: OOB MUTEX released
Sun Nov  6 13:23:38.394 2022 : 1995837456: SendOobPacket: Exit retVal=0x0
Sun Nov  6 13:23:38.395 2022 : 1995837456: SendOobDCMD: Exit dcmdOpCode=0x1020100 or controlCmd=0x0, retVal=0x0
Sun Nov  6 13:23:38.395 2022 : 1995837456: ProcessLibCommand: Exit: cmdType = 0x1, cmd = 0x1, retVal = 0x0
Sun Nov  6 13:23:38.396 2022 : 1995837456: ProcessLibCommand: Entry cmdType = 0x1, cmd = 0x2
Sun Nov  6 13:23:38.396 2022 : 1995837456: CSLSystem_GetCtrl: CSLSystem mutex acquired
Sun Nov  6 13:23:38.399 2022 : 1995837456: CSLSystem_GetCtrl: CSLSystem mutex released
Sun Nov  6 13:23:38.400 2022 : 1995837456: SL_ValidateInput: Controller Found [ctrlId] 0xa02
Sun Nov  6 13:23:38.400 2022 : 1995837456: CSLSystem_GetCtrl: CSLSystem mutex acquired
Sun Nov  6 13:23:38.401 2022 : 1995837456: CSLSystem_GetCtrl: CSLSystem mutex released
Sun Nov  6 13:23:38.402 2022 : 1995837456: ProcessLibCommand: Trying to acquire ctrl 0xa02 mutex
Sun Nov  6 13:23:38.402 2022 : 1995837456: ProcessLibCommand: ctrl 0xa02 mutex acquired
Sun Nov  6 13:23:38.403 2022 : 1995837456: ProcessLibCommand: Exit: cmdType = 0x1, cmd = 0x2, retVal = 0x800C <-------
-----error, should return 0x0
Sun Nov  6 13:23:38.411 2022 : 1995837456: ProcessLibCommand: Entry cmdType = 0x0, cmd = 0x9
Sun Nov  6 13:23:38.412 2022 : 1995837456: SL_CSLSystem_CleanUp: CSLSystem mutex acquired
Sun Nov  6 13:23:38.412 2022 : 1995837456: SL_CSLSystem_CleanUp: CSLSystem mutex released
```

5 Dbus API

Dbus APIs is provided for further development. For more information, please refer to the following document - ***Supervyse Storage System RAIDService DBus API***.

6 Redfish

6.1 Storage for RAID

6.1.1 Description

Storage defines a storage subsystem and its respective properties. A storage subsystem represents a set of storage controllers (physical or virtual) and the resources such as volumes that can be accessed from that subsystem.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1

6.1.2 Support Method and Privilege

	NoAuth	Login	ConfigureManager	ConfigureComponents	ConfigureUsers	ConfigureSelf
GET		V				

6.1.3 Properties

Property	Description
Drives	The set of drives attached to the storage controllers represented by this resource.
Volumes	The set of volumes produced by the storage controllers represented by this resource.
Status	See Common Properties *Status Object*.
StorageControllers	The set of storage controllers represented by this resource. See next section "StorageController for RAID" for more information.

6.1.3.1 Example

This example is the result of Storage schema's GET response. In the example, the SystemID is "system".


```
{
  "@odata.context": "/redfish/v1/$metadata#Storage.Storage",
  "@odata.etag": "a62cbe57b6f7ea3eaf0705e4e50275d8",
  "@odata.id": "/redfish/v1/Systems/system/Storage/1",
  "@odata.type": "#Storage.v1_8_0.Storage",
  "Description": "Storage System",
  "Drives": [
    {
      "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/0"
    },
    {
      "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/1"
```

```
        },
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/2"
        },
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/3"
        },
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/4"
        },
                    .
                    .
                    .
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/49"
        },
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/50"
        },
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/51"
        }
    ],
    "Id": "1",
    "Name": "Storage System",
    "Status": {
        "Health": "OK",
        "HealthRollup": "OK",
        "State": "Enabled"
    },
    "StorageControllers": [
        {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1#/StorageControllers/0",
            "@odata.type": "#Storage.v1_4_0.StorageController",
            "Actions": {
                "Oem": {
                    "#InsydeOEMExtensions.ClearConfiguration": {
                        "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ClearConfiguration"
                    },
                    "#InsydeOEMExtensions.ClearEvents": {
                        "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ClearEvents"
                    },
                    "#InsydeOEMExtensions.ClearForeignConfiguration": {
                        "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ClearForeignConfiguration"
                    },
                    "#InsydeOEMExtensions.CreateVolumeBasicData": {
                        "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.CreateVolumeBasicData"
                    },
                    "#InsydeOEMExtensions.CreateVolumeParameters": {
                        "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.CreateVolumeParameters"
                    },
                    "#InsydeOEMExtensions.GetFreeRaidResource": {
                        "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.GetFreeRaidResource"
                    },
                    "#InsydeOEMExtensions.ImportForeignConfiguration": {
```

```
                "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ImportForeignConfiguration"
            },
            "#InsydeOEMExtensions.ScanForeignConfiguration": {
                "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ScanForeignConfiguration"
            },
            "#InsydeOEMExtensions.StartPatrolRead": {
                "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.StartPatrolRead"
            },
            "#InsydeOEMExtensions.StopPatrolRead": {
                "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.StopPatrolRead"
            }
        }
    },
    "AssetTag": null,
    "FirmwareVersion": "5.170.01-3483",
    "Links": {
        "StorageServices": [
            {
                "@odata.id": "/redfish/v1/StorageServices/1"
            }
        ]
    },
    "Manufacturer": "Broadcom",
    "MemberId": "0",
    "Model": "AVAGO MegaRAID SAS 9440-8i",
    "Name": "Broadcom RAID/HBA Controller",
    "Oem": {
        "InsydeControllers": {
            "@odata.type": "#InsydeOEMExtensions.RaidControllers",
            "BBUCurrentMilliAmps": null,
            "BBUHealthPercent": null,
            "BBUName": null,
            "BBUPresent": false,
            "BBURemainingBatteryPercent": null,
            "BBUSerialNumber": null,
            "BBUStatus": null,
            "BBUTemperatureCelsius": null,
            "BBUType": null,
            "BBUVoltageMilliVolts": null,
            "BIOSVersion": "7.17.00.0_0x07110200",
            "Chip": "MegaRAID Tri-Mode SAS3408",
            "CreateRaidVolume": {
                "AccessPolicy": [],
                "DiskCachePolicy": [],
                "FreeDriveIds": [],
                "InitState": [],
                "MaxSize": 0,
                "PossibleRaidLevel": [],
                "StripeSize": []
            },
            "Events": {
                "@odata.id": "/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Oem/Insyde/Events"
            },
            "ExpanderFWVersion": null,
            "ExpanderTemp": null,
            "ForeignConfiguration": {
                "RaidLevel": "",
```

```
                "Size": 0,
                "State": "",
                "TargetId": ""
            },
            "HostInterface": "PCIe",
            "IoPolicySupport": false,
            "JunctionTemperature": 45,
            "Memory": 0,
            "MemoryChangeable": "No",
            "MemoryCorrectErrCount": 0,
            "MemoryUnCorrectErrCount": 0,
            "PCIDeviceID": "0x0017",
            "PCISubSystemID": "0x9440",
            "PCISubVersionID": "0x1000",
            "PCIVendorID": "0x1000",
            "PDList": [],
            "PackageVersion": "51.17.0-3971",
            "PortCount": 8,
            "SEEPROM": null,
            "SeqNumLastCleanShutdownEvent": 9,
            "SeqNumLastClearEvent": 2,
            "SeqNumNewestEvent": 213,
            "SeqNumOldestEvent": 1,
            "SeqNumThisSessionBootEvent": 208,
            "TTYLog": {
                "@odata.id": "/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Oem/Insyde/TTYLog"
            },
            "UEFIVersion": null
        }
    },
    "PartNumber": null,
    "SKU": null,
    "SerialNumber": "SP84924552",
    "SpeedGbps": 12,
    "Status": {
        "Health": "OK",
        "HealthRollup": "OK",
        "State": "Enabled"
    },
    "SupportedControllerProtocols": [
        "PCIe"
    ],
    "SupportedDeviceProtocols": [
        "SAS",
        "SATA"
    ]
},
{
    "@odata.id": "/redfish/v1/Systems/system/Storage/1#/StorageControllers/1",
    "@odata.type": "#Storage.v1_4_0.StorageController",
    "Actions": {
        "Oem": {}
    },
    "AssetTag": null,
    "FirmwareVersion": "14.250.14.00",
    "Links": {
        "StorageServices": [
            {
                "@odata.id": "/redfish/v1/StorageServices/1"
            }
        ]
```

```
        },
        "Manufacturer": "Broadcom",
        "MemberId": "1",
        "Model": "HBA 9500-8e",
        "Name": "Broadcom RAID/HBA Controller",
        "Oem": {
          "InsydeControllers": {
            "@odata.type": "#InsydeOEMExtensions.RaidControllers",
            "BBUCurrentMilliAmps": null,
            "BBUHealthPercent": null,
            "BBUName": null,
            "BBUPresent": false,
            "BBURemainingBatteryPercent": null,
            "BBUSerialNumber": null,
            "BBUStatus": null,
            "BBUTemperatureCelsius": null,
            "BBUType": null,
            "BBUVoltageMilliVolts": null,
            "BIOSVersion": "09.29.00.00_15.00.00.00",
            "Chip": "Fusion-MPT 12GSAS/PCIe Secure SAS38xx",
            "CreateRaidVolume": {
              "AccessPolicy": [],
              "DiskCachePolicy": [],
              "FreeDriveIds": [],
              "InitState": [],
              "MaxSize": 0,
              "PossibleRaidLevel": [],
              "StripeSize": []
            },
            "Events": {
              "@odata.id": "/redfish/v1/Systems/system/Storage/1/StorageControllers/1/Oem/Insyde/Events"
            },
            "ExpanderFWVersion": null,
            "ExpanderTemp": null,
            "ForeignConfiguration": {
              "RaidLevel": "",
              "Size": 0,
              "State": "",
              "TargetId": ""
            },
            "HostInterface": "PCIe",
            "IoPolicySupport": false,
            "JunctionTemperature": 42,
            "Memory": 0,
            "MemoryChangeable": "No",
            "MemoryCorrectErrCount": 0,
            "MemoryUnCorrectErrCount": 0,
            "PCIDeviceID": "0x00e6",
            "PCISubSystemID": "0x4080",
            "PCISubVersionID": "0x1000",
            "PCIVendorID": "0x1000",
            "PDList": [],
            "PackageVersion": "00.00.00.00",
            "PortCount": 11,
            "SEEPROM": null,
            "SeqNumLastCleanShutdownEvent": 0,
            "SeqNumLastClearEvent": 0,
            "SeqNumNewestEvent": 75,
            "SeqNumOldestEvent": 1,
            "SeqNumThisSessionBootEvent": 0,
            "UEFIVersion": null
```

```
            }
        },
        "PartNumber": null,
        "SKU": null,
        "SerialNumber": "SP934P0085",
        "SpeedGbps": 12,
        "Status": {
            "Health": "OK",
            "HealthRollup": "OK",
            "State": "Enabled"
        },
        "SupportedControllerProtocols": [
            "PCIe"
        ],
        "SupportedDeviceProtocols": [
            "SAS",
            "SATA"
        ]
    }
  ],
  "Volumes": {
    "@odata.id": "/redfish/v1/Systems/system/Storage/1/Volumes"
  }
}
```


6.2 StorageController for RAID

6.2.1 Description

Storage defines a storage subsystem and its respective properties. A storage subsystem represents a set of storage controllers (physical or virtual) and the resources such as volumes that can be accessed from that subsystem.
URI:
https://{ip}/redfish/v1/Systems/<SystemID>/Storage/<StorageId>#/StorageControllers/<StorageControllersId>

6.2.2 Support Method and Privilege

	NoAuth	Login	ConfigureManager	ConfigureComponents	Configure Users	ConfigureSelf
GET		V				
POST				V		

6.2.3 Properties

Property	Description
AssetTag	The asset tag for the controller.
Status	This type describes the status and health of a resource and its children.

SpeedGbps	The speed of the storage controller interface.
FirmwareVersion	The firmware version of this storage Controller.
Manufacturer	This is the manufacturer of this storage controller.
Model	This is the model number for the storage controller.
SKU	The SKU for this storage controller.
SerialNumber	The serial number for this storage controller.
PartNumber	The part number for this storage controller.
SupportedControllerProtocols	This represents the protocols by which this storage controller can be communicated to.
SupportedDeviceProtocols	This represents the protocols which the storage controller can use to communicate with attached devices.
Actions	The available actions for this Resource.
Oem	Oem extension object.
→ InsydeControllers	Oem extension object for storage controllers.
→→ SeqNumNewestEvent	The newest sequential number of event.
→→ SeqNumOldestEvent	The oldest sequential number of event.
→→ SeqNumLastClearEvent	The last clear sequential number of event.
→→ SeqNumLastCleanShutdownEvent	The last clear shutdown sequential number of event.
→→ SeqNumThisSessionBootEvent	This session boot sequential number of event.
→→ IoPolicySupport	This property if IO policy is supported.
→→ BBUPresent	The present of the BBU module.
→→ BBUStatus	The status of the BBU module.
→→ BBUName	The name of the BBU module.
→→ BBUType	The type of the BBU module.
→→ BBUTemperatureCelsius	The temperature of the BBU module.
→→ BBUVoltageMilliVolts	The Voltage(mV) of the BBU module. The temperature of the BBU module.
→→ BBUCurrentMilliAmps	The Current(mA) of the BBU module.
→→ BBUSerialNumber	The serial number of the BBU module.

→→ BBUHealthPercent	The health percentage of BBU module.
→→ BBURemainingBatteryPercent	The Remaining Battery percentage of the BBU module.
→→ Memory	This indicates the Memory Capacity.
→→ PackageVersion	This indicates the version of the controller package.
→→ JunctionTemperature	The temperature of the Storage Controller.
→→ PCIVendorID	This property shall contain the PCI vendor ID of the controller.
→→ PCIDeviceID	This property shall contain the PCI device ID of the controller.
→→ PCISubVersionID	This property shall contain the PCI sub version ID of the controller.
→→ PCISubSystemID	This property shall contain the PCI device ID of the controller.
→→ BIOSVersion	This property shall contain the BIOS version of the controller.
→→ UEFIVersion	This property shall contain the UEFI version of the controller.
→→ ExpanderFWVersion	This property shall contain the expander firmware version of the controller.
→→ SEEPROM	This property shall contain the SEEPROM of the controller.
→→ MemoryChangeable	This property shall contain the memory changeable of the controller.
→→ MemoryCorrectErrCount	This property shall contain the memory correct error count of the controller.
→→ MemoryUnCorrectErrCount	This property shall contain the memory uncorrect error count of the controller.
→→ Chip	This property shall contain the chip of the controller.
→→ ExpanderTemp	This property shall contain the expander temperature of the controller.
→→ HostInterface	This property shall contain the host interface of the controller.
→→ PortCount	This property shall contain the port count of the controller.

→→ TTYLog	The link to a collection of the TTY logs of the controller.
→→ Events	The link to a collection of the events logs of the controller.
→ CreateRaidVolume	This object indicates the allowable value for creating raid volume.
→→ MaxSize	The max size to create raid volume.
→→ PossibleRaidLevel	The Possible Raid Level to create raid volume.
→→ FreeDriveIds	The Free Drive Ids to create raid volume.
→→ StripeSize	The Stripe Size to create raid volume.
→→ AccessPolicy	The Access Policy to create raid volume.
→→ DiskCachePolicy	The Disk Cache Policy to create raid volume.
→→ InitState	The Init State to create raid volume.
→ ForeignConfiguration	This object indicates the allowable value for Foreign Configuration.
→→ TargetId	The Target Id of foreign configuration.
→→ RaidLevel	The Raid Level of foreign configuration.
→→ Size	The Size of foreign configuration.
→→ State	The State of foreign configuration.
→ PDList	The physical disk list of this controller.
→→ DeviceId	The device Id of physical disk.
→→ EnclosureDeviceId	The enclosure device Id of physical disk.
→→ SlotNumber	The SlotNumber of physical disk.
→→ ConnectedPortBitmap	The connected port bitmap of physical disk.
→→ SASAddress	The SAS address of foreign configuration.

6.2.3.1 Example

This example is the result of StorageControllers GET response. In this example, the SystemID is "system".


```
{
    "@odata.id": "/redfish/v1/Systems/system/Storage/1#/StorageControllers/0",
    "@odata.type": "#Storage.v1_4_0.StorageController",
    "Actions": {
        "Oem": {
            "#InsydeOEMExtensions.ClearConfiguration": {
                "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ClearConfiguration"
```

```
        },
        "#InsydeOEMExtensions.ClearEvents": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ClearEvents"
        },
        "#InsydeOEMExtensions.ClearForeignConfiguration": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ClearForeignConfiguration"
        },
        "#InsydeOEMExtensions.CreateVolumeBasicData": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.CreateVolumeBasicData"
        },
        "#InsydeOEMExtensions.CreateVolumeParameters": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.CreateVolumeParameters"
        },
        "#InsydeOEMExtensions.GetFreeRaidResource": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.GetFreeRaidResource"
        },
        "#InsydeOEMExtensions.ImportForeignConfiguration": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ImportForeignConfiguration"
        },
        "#InsydeOEMExtensions.ScanForeignConfiguration": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.ScanForeignConfiguration"
        },
        "#InsydeOEMExtensions.StartPatrolRead": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.StartPatrolRead"
        },
        "#InsydeOEMExtensions.StopPatrolRead": {
          "target":
"/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Actions/Oem/InsydeOEMExtensions.StopPatrolRead"
        }
      }
    },
    "AssetTag": null,
    "FirmwareVersion": "5.170.01-3483",
    "Links": {
      "StorageServices": [
        {
          "@odata.id": "/redfish/v1/StorageServices/1"
        }
      ]
    },
    "Manufacturer": "Broadcom",
    "MemberId": "0",
    "Model": "AVAGO MegaRAID SAS 9440-8i",
    "Name": "Broadcom RAID/HBA Controller",
    "Oem": {
      "InsydeControllers": {
        "@odata.type": "#InsydeOEMExtensions.RaidControllers",
        "BBUCurrentMilliAmps": null,
        "BBUHealthPercent": null,
        "BBUName": null,
        "BBUPresent": false,
        "BBURemainingBatteryPercent": null,
        "BBUSerialNumber": null,
```

```
        "BBUStatus": null,
        "BBUTemperatureCelsius": null,
        "BBUType": null,
        "BBUVoltageMilliVolts": null,
        "BIOSVersion": "7.17.00.0_0x07110200",
        "Chip": "MegaRAID Tri-Mode SAS3408",
        "CreateRaidVolume": {
            "AccessPolicy": [],
            "DiskCachePolicy": [],
            "FreeDriveIds": [],
            "InitState": [],
            "MaxSize": 0,
            "PossibleRaidLevel": [],
            "StripeSize": []
        },
        "Events": {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Oem/Insyde/Events"
        },
        "ExpanderFWVersion": null,
        "ExpanderTemp": null,
        "ForeignConfiguration": {
            "RaidLevel": "",
            "Size": 0,
            "State": "",
            "TargetId": ""
        },
        "HostInterface": "PCIe",
        "IoPolicySupport": false,
        "JunctionTemperature": 45,
        "Memory": 0,
        "MemoryChangeable": "No",
        "MemoryCorrectErrCount": 0,
        "MemoryUnCorrectErrCount": 0,
        "PCIDeviceID": "0x0017",
        "PCISubSystemID": "0x9440",
        "PCISubVersionID": "0x1000",
        "PCIVendorID": "0x1000",
        "PDList": [],
        "PackageVersion": "51.17.0-3971",
        "PortCount": 8,
        "SEEPROM": null,
        "SeqNumLastCleanShutdownEvent": 9,
        "SeqNumLastClearEvent": 2,
        "SeqNumNewestEvent": 213,
        "SeqNumOldestEvent": 1,
        "SeqNumThisSessionBootEvent": 208,
        "TTYLog": {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/StorageControllers/0/Oem/Insyde/TTYLog"
        },
        "UEFIVersion": null
    }
},
"PartNumber": null,
"SKU": null,
"SerialNumber": "SP84924552",
"SpeedGbps": 12,
"Status": {
    "Health": "OK",
    "HealthRollup": "OK",
    "State": "Enabled"
},
```

```
  "SupportedControllerProtocols": [
    "PCIe"
  ],
  "SupportedDeviceProtocols": [
    "SAS",
    "SATA"
  ]
}
```


6.2.4 Supported Actions

15. **GetFreeRaidResource**

Description: This action is used to get free raid resouce.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.GetFreeRaidResource

Property	Description
N/A	N/A

16. CreateVolumeBasicData

Description: This action is used to active the first stage to create RAID.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.CreateVolumeBasicData

Property	Description
RaidLevel	Selected Raid Level.
SelectedDriveIds	Selected Drive Ids.

Example Payload


```
{"RaidLevel" : "1", " SelectedDriveIds " : [ "1", "2" ] }
```


17. CreateVolumeParameters

Description: This action is used to create volume parameters.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.CreateVolumeParameters

Property	Allowed Value	Description
Name	STRING	Volume Name.
Size	NUMBER	Volume Size.
StripeSize (KB)	16, 32, 64, 128, 256,	The Stripe Size to create raid volume.

Property	Allowed Value	Description
	512, 1024	
AccessPolicy	"Read Write"	Set access policy to read write.
	"Read Only"	Set access policy to read only.
	"Blocked"	Set access policy to blocked.
DiskCachePolicy	"Unchanged"	Set disk cache policy to Unchanged.
	"Enable"	Set disk cache policy to enabled.
	"Disable"	Set disk cache policy to disabled.
InitState	"No Init"	Set init state to no init.
	"Quick Init"	Set init state to quick init.
	"Full Init"	Set init state to full init.

Example Payload


```
{
"Name":"VolumeName",
"Size":237952,
"StripeSize":256,
"AccessPolicy":"Read Only",
"DiskCachePolicy":"Enable",
"InitState":"No Init"
}
```


18. ScanForeignConfiguration

Description: This action is used to scan foreign configuration.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.ScanForeignConfiguration

Property	Description
N/A	N/A

19. ImportForeignConfiguration

Description: This action is used to import foreign configuration.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.ImportForeignConfiguration

Property	Description
N/A	N/A

20. **ClearForeignConfiguration**

Description: This action is used to clear foreign configuration.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.ClearForeignConfiguration

Property	Description
N/A	N/A

21. StartPatrolRead

Description: This action is used to start patrol read.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.StartPatrolRead

Property	Description
N/A	N/A

22. StopPatrolRead

Description: This action is used to stop patrol read.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.StopPatrolRead

Property	Description
N/A	N/A

23. ClearConfiguration

Description: This action is used to clear configuration.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/StorageControllers/<StorageControllers ID>/Actions/Oem/InsydeOEMExtensions.ClearConfiguration

Property	Description
N/A	N/A

24. **Clear Events**

Description: This action is used to clear events of the storage controller.
URI:
https://{ip}/redfish/v1/Systems/<SystemID>/Storage/<StorageId>/StorageControllers/<StorageControllerId>/Actions/Oem/InsydeOEMExtensions.ClearEvents

Property	Description
N/A	N/A

6.3 Drive for RAID

6.3.1 Description

The Drive schema represents a single physical disk drive for a system, including links to associated Volumes.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/<StorageId>/Drives/<DriveId>

URI: https://{ip}/redfish/v1/StorageServices/1/Drives/<DriveId>

6.3.2 Support Method and Privilege

	NoAuth	Login	Configure Manager	ConfigureComponents	Configure Users	ConfigureSelf
GET		V				
POST				V		

6.3.3 Properties

Note: Property shows null if device was unable to obtain.

Property	Description
Model	This is the model number for the drive.
SerialNumber	The serial number for this drive.
Revision	The revision of this Drive. This is typically the firmware/hardware version of the drive.
Protocol	The protocol this drive is using to communicate to the storage controller.
MediaType	The type of media contained in this drive.
Manufacturer	This is the manufacturer of this drive.
SKU	This is the SKU for this drive.
PartNumber	The part number for this drive.
Status	See Common Properties *Status Object*.
CapacityBytes	The size in bytes of this Drive.

FailurePredicted	Is this drive currently predicting a failure in the near future.
HotspareType	The type of hotspare this drive is currently serving as.
EncryptionAbility	The encryption abilities of this drive.
EncryptionStatus	The status of the encrytion of this drive.
BlockSizeBytes	The size of the smallest addressible unit (Block) of this drive in bytes.
CapableSpeedGbs	The speed which this drive can communicate to a storage controller in ideal conditions in Gigabits per second.
NegotiatedSpeedGbs	The speed which this drive is currently communicating to the storage controller in Gigabits per second.
RotationSpeedRPM	The rotation speed of this Drive in Revolutions per Minute (RPM).
PredictedMediaLifeLeftPercent	The percentage of reads and writes that are predicted to still be available for the media.
StatusIndicator	The state of the status indicator, used to communicate status information about this drive.
IndicatorLED	The state of the indicator LED, used to identify the drive.
Actions	The available actions for this Resource.
Oem	Oem extension object.
→ InsydeDrives	Oem extension object for Drives.
→→ CopyBackProgressPercent	The copy back progress percent
→→ PatrolReadProgressPercent	The patrol read progress percent
→→ SlotNumber	The slot number of drive.
→→ StorageControllerReference	This indicates the reference storagecontroller index.
→→ DriveState	The state of standbyoffline drive.
→→ RawId	This indicates the drive index.
→→ AddtionalState	This indicates the additional state of drive.
→→ SMART	The oem extension object for SMART Reading.
→→→ Available	This indicates if the SAMRT reading is available.

→→→ Awake	This indicates if the Awake operation is available.
→→→ HealthGood	This indicates the health status.
→→→ OfflineDataCollectionStatus	This indicates the off line Data Collection status.
→→→ TotalOfflineDataCollectionSeconds	This indicates the Total off line Data Collection Seconds.
→→→ SelfTestExecutionStatus	This indicates the Self Test Execution Status.
→→→ SelfTestExecutionPercentRemaining	This indicates the Self Test Execution Percent Remaining.
→→→ ConveyanceTestAvailable	This indicates if the Awake operation is available.
→→→ ShortAndExtendedTestAvailable	This indicates if the Awake operation is available.
→→→ StartTestAvailable	This indicates if the Awake operation is available.
→→→ AbortTestAvailable	This indicates if the Awake operation is available.
→→→ ShortTestPollingMinutes	This indicates the short test polling minutes.
→→→ ExtendedTestPollingMinutes	This indicates the extended test polling minutes.
→→→ ConveyanceTestPollingMinutes	This indicates the conveyance test polling minutes.
→→→ BadSectors	This indicates the conveyance test polling minutes.
→→→ PowerOn	The value of this property shall be the amount of time the Memory module was powered on during its lifetime.
→→→ PowerCycle	The value of this property shall be number of power cycles over the lifetime.
→→→ RPM	This indicates the RPM status.
→→→ OverallHealth	This indicates the drive overall health.
→→→ Temperature	The temperature of drive.
→→→ Attribute	The object for smart attributes.
→→→→ SmartAttributes[]	The smart attributes by index if support MCTP
Links	Contains references to other resources that are related to this resource.

→ Volumes	An array of references to the volumes contained in this drive. This will reference Volumes that are either wholly or only partly contained by this drive.

6.3.3.1 Example

This example is the result of Drive for Raid's GET response.


```
{
  "@odata.context": "/redfish/v1/$metadata#Drive.Drive",
  "@odata.etag": "45dada377c136679a879d597cb4342c6",
  "@odata.id": "/redfish/v1/StorageServices/1/Drives/0",
  "@odata.type": "#Drive.v1_9_0.Drive",
  "Actions": {
    "Oem": {
      "#InsydeOEMExtensions.StartLocatePD": {
        "target": "/redfish/v1/StorageServices/1/Drives/0/Actions/Oem/InsydeOEMExtensions.StartLocatePD"
      },
      "#InsydeOEMExtensions.StopLocatePD": {
        "target": "/redfish/v1/StorageServices/1/Drives/0/Actions/Oem/InsydeOEMExtensions.StopLocatePD"
      }
    }
  },
  "BlockSizeBytes": 0,
  "CapableSpeedGbs": null,
  "CapacityBytes": null,
  "Description": null,
  "EncryptionAbility": null,
  "EncryptionStatus": null,
  "FailurePredicted": null,
  "HotspareType": null,
  "Id": "0",
  "IndicatorLED": null,
  "Manufacturer": null,
  "MediaType": null,
  "Model": null,
  "Name": "Physical Disk 0",
  "NegotiatedSpeedGbs": null,
  "Oem": {
    "InsydeDrives": {
      "@odata.type": "#InsydeOEMExtensions.RaidDrive",
      "AddtionalState": null,
      "CopyBackProgressPercent": null,
      "DriveState": null,
      "PatrolReadProgressPercent": null,
      "RawId": null,
      "SMART": {
        "AbortTestAvailable": null,
        "Attribute": null,
        "Available": null,
        "Awake": null,
        "BadSectors": null,
        "ConveyanceTestAvailable": null,
        "ConveyanceTestPollingMinutes": null,
        "ExtendedTestPollingMinutes": null,
        "HealthGood": null,
        "OfflineDataCollectionStatus": null,
        "OverallHealth": null,
        "PowerCycle": null,
```

```
            "PowerOn": null,
            "RPM": null,
            "SelfTestExecutionPercentRemaining": null,
            "SelfTestExecutionStatus": null,
            "ShortAndExtendedTestAvailable": null,
            "ShortTestPollingMinutes": null,
            "StartTestAvailable": null,
            "Temperature": null,
            "TotalOfflineDataCollectionSeconds": null
        },
        "SlotNumber": null,
        "StorageControllerReference": 0
    }
  },
  "PartNumber": null,
  "PredictedMediaLifeLeftPercent": null,
  "Protocol": null,
  "Revision": null,
  "RotationSpeedRPM": null,
  "SKU": null,
  "SerialNumber": null,
  "Status": {
    "State": "Absent"
  },
  "StatusIndicator": null
}
```


6.3.4 Supported Actions

25. OEM Make Dedicated Hotspare

Description: This action is used to make dedicated hotspare for specific volume.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.MakeDedicatedHotspare

Property	Description
VolumeID	Indicates the Volume ID.

Example Payload

```
{"VolumeID":"Broadcom10"}
```

26. OEM MakeGlobalHotspare

Description: This action is used to make global hotspare for all system.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.MakeGlobalHotspare

Property	Description
N/A	N/A

27. OEM Remove Hotspare

Description: This action is used to remove hotspare.

URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.RemoveHotspare

Property	Description
N/A	N/A

28. **OEM Start Replace PD**

Description: This action is used to replace the drive.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.StartReplacePD

Property	Description
DriveID	Indicates the Drive ID.

Example Payload

```
{"DriveID":"1"}
```

29. **OEM Stop Replace PD**

Description: This action is used to stop replacing the drive.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.StopReplacePD

Property	Description
N/A	N/A

30. **OEM Start Locate PD**

Description: This action is used to start locating the drive.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.StartLocatePD

Property	Description
N/A	N/A

31. **OEM Stop Locate PD**

Description: This action is used to stop locating the drive.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/1/Drives/<Drive ID>/Action/Oem/InsydeOEMExtensions.StopLocatePD

Property	Description
N/A	N/A

6.4 VolumeCollection for RAID

Refs: VolumeCollection.VolumeCollection

6.4.1 Description

A Collection of Storage resource instances.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/<StorageId>/Volumes

URI: https://{ip}/redfish/v1/StorageServices/<StorageServicesId>/Volumes

6.4.2 Support Method and Privilege

	NoAuth	Login	ConfigureManager	ConfigureComponents	ConfigureUsers	ConfigureSelf
GET		V				

6.4.3 Properties

For property table of Collection type Schema, please refer to *Collection Resources* Section.

6.4.3.1 Example

This example is the result of VolumeCollection schema's GET response.

```
{
   "@odata.context": "/redfish/v1/$metadata#VolumeCollection.VolumeCollection",
   "@odata.type": "#VolumeCollection.VolumeCollection",
   "@odata.id": "/redfish/v1/StorageServices/1/Volumes",
   "Name": "Storage Volume Collection",
   "Members@odata.count": 0,
   "Members": [],
   "@odata.etag": "c120ee65badad1d6185e829a72ad7cd1"
}
```

6.5 Volume for RAID

Refs: Volume.v1_6_2.Volume

6.5.1 Description

Volume contains properties used to describe a volume, virtual disk, LUN, or other logical storage entity for any system.
URI: https://{ip}/redfish/v1/Systems/<SystemID>/Storage/<StorageId>/Volumes/<VolumeId>

URI: https://{ip}/redfish/v1/StorageServices/<StorageServicesId>/Volumes/<VolumeId>

6.5.2 Support Method and Privilege

	NoAuth	Login	Configure Manager	ConfigureComponents	Configure Users	ConfigureSelf

GET		V				
PATCH				V		
POST				V		
DELETE				V		

6.5.3 Properties

Property	**Description**
Status	See Common Properties *Status Object*.
CapacityBytes	The size in bytes of this Drive.
VolumeType	The type of this volume.
Encrypted	Is this Volume encrypted.
BlockSizeBytes	The size of the smallest addressible unit (Block) of this drive in bytes.
Links	Contains references to other resources that are related to this resource.
→ Drives	An array of references to the drives which contain this volume. This will reference Drives that either wholly or only partly contain this volume.
Operations	An operation currently running on this resource.
→ OperationName	The name of the operation.
→ PercentageComplete	The percentage of the operation that has been completed.
OptimumIOSizeBytes	The size in bytes of this Volume's optimum IO size.
Actions	The available actions for this Resource.
Oem	Oem extension object.
→ InsydeVolume	Oem extension object for volume..
→→ RaidLevel	The raid level of the volume.
→→ State	The state of the volume.
→→ ReadPolicy	The read policy of the volume.
→→ WritePolicy	The write policy of the volume.
→→ IoPolicy	The IO policy of the volume.

→→ AccessPolicy	The access policy of the volume.
→→ DiskCachePolicy	The disk cache policy of the volume.
→→ StripSize	The strip size of the volume.
→→ BadBlocks	Check whether the volume has bad blocks.
→→ StorageControllerReference	This indicates the reference storagecontroller index.

6.5.3.1 Example

This example is the result of Volume GET response.


```
{
   "@odata.context": "/redfish/v1/$metadata#Volume.Volume",
   "@odata.etag": "47e8369d8a1e8b402c61d197266a4c33",
   "@odata.id": "/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1",
   "@odata.type": "#Volume.v1_5_0.Volume",
   "Actions": {
      "#Volume.ChangeRAIDLayout": {
         "target": "/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1/Actions/Volume.ChangeRAIDLayout"
      },
      "#Volume.CheckConsistency": {
         "target": "/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1/Actions/Volume.CheckConsistency"
      },
      "Oem": {
         "#InsydeOEMExtensions.CancelCheckConsistency": {
            "target":
"/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1/Actions/Oem/InsydeOEMExtensions.CancelCheckConsistency"
         },
         "#InsydeOEMExtensions.ChangeLDConfig": {
            "target":
"/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1/Actions/Oem/InsydeOEMExtensions.ChangeLDConfig"
         },
         "#InsydeOEMExtensions.StartLocateLD": {
            "target":
"/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1/Actions/Oem/InsydeOEMExtensions.StartLocateLD"
         },
         "#InsydeOEMExtensions.StopLocateLD": {
            "target":
"/redfish/v1/Systems/system/Storage/1/Volumes/Broadcom0_1/Actions/Oem/InsydeOEMExtensions.StopLocateLD"
         }
      }
   },
   "BlockSizeBytes": 512,
   "CapacityBytes": 249510756352,
   "Description": "Logical Disk Broadcom0_1",
   "Encrypted": false,
   "Id": "Broadcom0_1",
   "Links": {
      "Drives": [
         {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/1"
         },
         {
            "@odata.id": "/redfish/v1/Systems/system/Storage/1/Drives/2"
         }
      ]
```

```
  },
  "Name": "Test",
  "Oem": {
    "InsydeVolume": {
      "@odata.type": "#InsydeOEMExtensions.RaidVolume",
      "AccessPolicy": "Read Write",
      "BadBlocks": false,
      "DiskCachePolicy": "Enable",
      "IoPolicy": "Direct IO",
      "RaidLevel": "1",
      "ReadPolicy": "No Read Ahead",
      "State": "Optimal",
      "StorageControllerReference": 0,
      "StripSize": 64,
      "WritePolicy": "Write Through"
    }
  },
  "Operations": [
    {
      "OperationName": null,
      "PercentageComplete": null
    }
  ],
  "OptimumIOSizeBytes": 512,
  "RAIDType": "1",
  "Status": {
    "Health": "OK",
    "State": "Enabled"
  },
  "StripSizeBytes": 64
}
```


6.5.4 Updatable Properties

Property	Allowed value	Description
ReadPolicy	"Read Ahead"	Reads sequential sectors of the virtual disk.
	"No Read Ahead"	Disable read ahead.
WritePolicy	"Write Through"	Send a write-request completion signal only after the data is written to the disk.
	"Write Back"	Send a write-request completion signal as soon as the data is in the controller cache but has not yet been written to disk.
	"Write Cached"	writes data to the write cache before writing data to the physical disk.
IoPolicy	"Direct IO"	The reads are not buffered in cache memory.

	"Cached IO"	The reads are buffered in cache memory.
AccessPolicy	"Read Write"	Data is readable and writable.
	"Read Only"	Data is read only.
	"Blocked"	Data is inaccessible.
DiskCachePolicy	"Unchanged"	Leave Disk Cache Policy unchanged.
	"Enable"	Disk Cache Policy is enabled.
	"Disable"	Disk Cache Policy is disabled.

6.5.4.1 Example Payload


```
{
"Oem" : {
"InsydeVolume" : {
"ReadPolicy" : "Read Ahead",
"WritePolicy" : "Write Through",
"IoPolicy" : "Direct IO",
"AccessPolicy" : "Read Write",
"DiskCachePolicy" : "Enable"
}
}
}
```


6.5.5 Supported Actions

32. **Check Consistency**

Description: This action is used to force a check of the Volume's parity or redundant data to ensure it matches calculated values.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/Volumes/<VOLUME ID>/Actions/Volume.CheckConsistency

Property	Description
N/A	N/A

33. **OEM Cancel Check Consistency**

Description: This action I used to cancel consistency check for volume.

URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/Volumes/<VOLUME ID>/Actions/Oem/InsydeOEMExtensions.CancelCheckConsistency

Property	Description
N/A	N/A

34. **OEM Start Locate LD**

Description: This action is used to start locate LD.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/Volumes/<VOLUME ID>/Actions/Oem/InsydeOEMExtensions.StartLocateLD

Property	Description
N/A	N/A

35. **OEM Stop Locate LD**

Description: This action is used to stop locate LD.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/Volumes/<VOLUME ID>/Actions/Oem/InsydeOEMExtensions.StopLocateLD

Property	Description
N/A	N/A

36. **OEM Change LD Config**

Description: This action is used to change a Virtual Drive Configuration.
URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/Volumes/<VOLUME ID>/Actions/Oem/InsydeOEMExtensions.ChangeLDConfig

Property	Allowed Value	Description
RaidLevel	"0"	Striped array with no parity.
	"1"	Mirrored array.
	"5"	Striped array with rotating parity, optimized for short, multi-threaded transfers.
	"6"	Similar to RAID-5, but with dual rotating parity physical disks, toleratingtwo physical disk failures.
SelectedDriveIds[]	STRING	Indicates the drive IDs.

Example Payload

```
{"RaidLevel":"1", "SelectedDriveIds" : ["1",”2”] }
```

6.5.6 Remove Volume by DELETE

Description: Volume can be removed by Deleting the Volume resource.

URI: https://{ip}/redfish/v1/StorageServices/<STORAGESERVICE ID>/Volumes/<VOLUME ID>

URI: https://{ip}/redfish/v1/Systems/<SYSYEM ID>/Storage/<STORAGE ID>/Volumes/<VOLUME ID>

7 IPMI

7.1 Get Storage Collection Info

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 01h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Storage Collection Info = 00h
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x1 0x0
 00
```

7.2 Get Storage Collection Info (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 01h 01h - StorageCollection 02h - Storage 03h - Controller

	byte	data field
		04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Storage Collection Info (Read) = 01h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2:33	Description
	34:65	Name
	66	Count
	67:70	Member(s)

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x1 0x1
 00 53 74 6f 72 61 67 65 20 53 79 73 74 65 6d 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 49 6e 73 79 64 65 20 53 74 6f 72 61 67 65 20
 53 79 73 74 65 6d 00 00 00 00 00 00 00 00 00 00
 00 01 01 00 00 00
```

7.3 Get Storage Info

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 02h 01h - StorageCollection 02h - Storage 03h - Controller

	byte	data field
		04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Storage Info = 00h
	5	Storage ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x2 0x0 0x1
 00
```

7.4 Get Storage Info (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 02h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Storage Info (Read) = 01h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed

	byte	data field
		03h - not support 04h - wrong parameters
	2:33	Description
	34:65	Name
	66	Controller Count
	67:74	Controller ID(s)
	75	Drive Count
	76:202	Drive ID(s)

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x2 0x1
 00 53 74 6f 72 61 67 65 20 53 79 73 74 65 6d 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 53 74 6f 72 61 67 65 20 53 79 73 74 65 6d 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 01 01 00 00 00 00 00 00 00 05 00 01 02 03 04
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00
```

7.5 Get Controller Info

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume

	byte	data field
	4	Get Controller Info = 00h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x0 0x1
 00
```

7.6 Get Controller Info (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Controller Info (Read) = 01h
	5	Controller ID = 01h 1 base
Response Data (LS byte first)		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support

	byte	data field
		04h - wrong parameters
	2:33	Firmware Version
	34:65	Package Version
	66:97	Manufacturer
	98:161	Model
	162:193	Serial Number
	194:225	BIOS Version
	226:257	UEFI Version
	258:289	Expander FW Version
	290:297	Health
	298:301	Memory Size
	302:303	PCI Vendor ID
	304:305	PCI Device ID
	306:307	PCI Subversion ID
	308:309	PCI Sybsystem ID
	310:311	Memory Correct Count
	312:313	Memory Uncorrect Count
	314	Junction Temperature
	315	Expander Temperature
	316	Port Count
	317	I2C bus
	318	MUX addr
	319	MUX data
	320	Controller index

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x1 0x1
 00 35 2e 31 37 30 2e 30 30 2d 33 34 38 33 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 35 31 2e 31 37 2e 30 2d 33 39 37 31 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 42 72 6f 61 64 63 6f 6d 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 41 56 41 47 4f 20 4d 65 67 61 52 41 49 44 20
 53 41 53 20 39 34 36 30 2d 31 36 69 00 00 00 00
```

```
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 53 50 37 33 34 33 37 36 31 33 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 37 2e 31 37 2e 30 30 2e 30 5f 30 78 30 37 31
31 30 32 30 30 00 00 00 00 00 00 00 00 00 00 00
00 4e 2f 41 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 4e 2f 41 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 4f 4b 00 00 00 00 00 00 00 10 00 00 00 10 14
00 00 10 60 94 00 00 00 00 28 ff 10 02 e0 04 01
```

7.7 Delete Logical Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Delete Logical Drive = 02h
	5	Controller ID = 01h 1 base
	6:37	Volume Name
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x2 0x1 0x42 0x72 0x6f 0x61 0x64
0x63 0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.8 Delete Logical Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Delete Logical Drive (Read) = 03h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x3
 00
```

7.9 Get Free Raid Resource

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection

	byte	data field
		02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Free Raid Resource = 04h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x4 0x1
 00
```

7.10 Get Free Raid Resource (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Free Raid Resource (Read) = 05h
Response Data		
	1	Completion Code

	byte	data field
	2:9	Support Raid Level(s)
	10:41	Drive ID(s)

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x5
 00 00 01 05 06 0a 00 00 00 04 05 06 07 03 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00
```

7.11 Create Volume Basic Data

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Create Volume Basic Data = 06h
	5	Controller ID = 01h 1 base
	6	Raid Level 0 1 5 6 10 50 60

	byte	data field
	7	Selected Drive Ids Count(s)
	8:~	Selected Drive Ids
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x6 0x1 0x1 0x2 0x4 0x5
00
```

7.12 Create Volume Basic Data (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Create Volume Basic Data (Read) = 07h
Response Data (LS byte first)		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2:5	Max Size

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x7
 00 80 a1 03 00
```

7.13 Create Volume Parameters

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Create Volume Parameters = 08h
	5	Controller ID = 01h 1 base
	6:9	Size
	10	Stripe Size 0x0 - 16 0x1 - 32 0x2 - 64 0x3 - 128 0x4 - 256 0x5 - 512 0x6 - 1024
	11	AccessPolicy 0x0 - Read Write 0x1 - Read Only 0x2 - Blocked

	byte	data field
	12	DiskCachePolicy 0x0 - Unchanged 0x1 - Enable 0x2 - Disable
	13	InitState 0x0 - No Init 0x1 - Quick Init 0x2 - Full Init
	14:45	Name
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x8 0x1 0x80 0xa1 0x3 0x0 0x2 0x0
0x0 0x0 0x74 0x65 0x73 0x74
 00
```

7.14 Create Volume Parameters (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Create Volume Parameters (Read) = 09h
Response Data		

	byte	data field
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x9
 00
```

7.15 Save TTY Log

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Save TTY Log = 0Ah
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0xA 0x1
 00
```

7.16 Save TTY Log (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h

	byte	data field
		01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Save TTY Log (Read) = 0Bh
Response Data	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0xB
 00
```

7.17 Scan Foreign Configuration

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume

	byte	data field
	4	Scan Foreign Configuration = 0Ch
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0xC 0x1
 00
```

7.18 Scan Foreign Configuration (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Scan Foreign Configuration (Read) = 0Dh
Response Data		
	1	Completion Code 00h - success

	byte	data field
		01h - running 02h - failed 03h - not support 04h - wrong parameters
	2	Target ID
	3	Raid Level
	4	State 0x0 - OFFLINE 0x1 - PARTIALLY DEGRADED 0x2 - DEGRADED 0x3 - OPTIMAL
	5:8	Size

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0xD
 00 01 01 02 80 a1 03 00
```

7.19 Import Foreign Configuration

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Import Foreign Configuration = 0Eh
	5	Controller ID = 01h 1 base

	byte	data field
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x0E 0x1
 00
```

7.20 Import Foreign Configuration (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Import Foreign Configuration (Read) = 11h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x0F
 00
```

7.21 Clear Foreign Configuration

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Clear Foreign Configuration = 10h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x10 0x1
 00
```

7.22 Clear Foreign Configuration (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller

	byte	data field
		04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Clear Foreign Configuration (Read) = 11h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x11
 00
```

7.23 Start Firmware Upddate

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume

	byte	data field
	4	Start Firmware Upddate = 12h Controller ID = 01h
	5	1 base
	6:37	location of firmware image with name
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x12 0x1 0x2f 0x74 0x6d 0x70 0x2f
0x39 0x34 0x36 0x30 0x2d 0x38 0x69 0x2e 0x72 0x6f 0x6d 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.24 Start Firmware Upddate (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Start Firmware Upddate (Read) = 13h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed

	byte	data field
		03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x13
 00
```

7.25 Start Patrol Read

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Start Patrol Read = 14h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x14 0x1
 00
```

7.26 Start Patrol Read (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)

	byte	data field
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Start Patrol Read (Read) = 15h
Response Data	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x15
 00
```

7.27 Stop Patrol Read

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event

	byte	data field
		05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop Patrol Read = 16h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x16 0x1
 00
```

7.28 Stop Patrol Read (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop Patrol Read (Read) = 17h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed

	byte	data field
		03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x17
 00
```

7.29 Delete all volume(s)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Delete all volume(s) = 18h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x18 0x1
 00
```

7.30 Delete all volume(s) (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h

	byte	data field
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Delete all volume(s) (Read) = 19h
Response Data	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x19
 00
```

7.31 Clear all logical event(s)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive

	byte	data field 06h - VolumeCollection 07h - Volume
	4	Clear all logical event(s) = 1Ah
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x1A 0x1
 00
```

7.32 Clear all logical event(s) (Read)

	Byte	Data Field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Clear all logical event(s) (Read) = 1Bh
Response Data	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x1B
 00
```

7.33 Set controller id for HPM firmware update

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 03h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Set Firmware Update Param For Controller = 1Ch
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x3 0x1C 0x1
 00
```

7.34 Get Event Sequence Information

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 04h 01h - StorageCollection

	byte	data field
		02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Event Sequence Information = 00h
	5	Controller ID = 01h 1 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x4 0x0 0x1
 00
```

7.35 Get Event Sequence Information (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 04h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Event Sequence Information (Read) = 01h
Response Data (LS byte first)		
	1	Completion Code

	byte	data field
		00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2:5	Newest Event
	6:9	Oldest Event
	10:13	Last Clear Event
	14:17	Last Clean Shutdown Event
	18:21	This Session Boot Event

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x4 0x1
 00 56 7f 01 00 ed 73 00 00 02 7f 01 00 6e 68 01
 00 30 7f 01 00
```

7.36 Get Event Details

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 04h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Event Details = 02h
	5	Controller ID = 01h 1 base
	6-9	SeqNumber (LS byte first)
Response Data		

	byte	data field
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x4 0x2 0x1 0x0e 0x7e 0x01 0x00
 00
```

7.37 Get Event Details (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 04h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Event Details (Read) = 03h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2:5	Sequence Number
	6	Level 0x0 - Debug 0x1 - Progress 0x2 - Information 0x3 - Warning

	byte	data field
		0x4 - Critical 0x5 - Fatal 0x6 - Dead 0x7 - Undifined
	7:38	Date Time
	39:102	Description

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x4 0x3
 00 bd 5d 00 00 02 31 2f 31 2f 32 30 31 36 20 3b
 20 30 3a 33 39 3a 35 39 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 74 61 74 65 20 63 68 61 6e
 67 65 20 6f 6e 20 50 44 20 31 62 28 65 30 78 38
 36 2f 73 30 29 20 66 72 6f 6d 20 4f 4e 4c 49 4e
 45 28 31 38 29 20 74 6f 20 55 4e 43 4f 4e 46 49
 47 55 52 45 44 5f
```

7.38 Get Drive Info

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Drive Info = 00h
	5	Drive ID = 00h 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x0 0x4
 00
```

7.39 Get Drive Info (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Drive Info (Read) = 01h
	5	Drive ID = 00h 0 base
Response Data	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2:33	Model
	34:65	Serial Number
	66:97	Revision
	98:105	Health
	106:113	Capacity
	114:115	RPM

byte	data field
116	State 0x0 - Absent 0x1 - StandbyOffline 0x2 - StandbyOffline (Foreign) 0x3 - StandbyOffline (BootSector) 0x4 - UnavailableOffline 0x5 - StandbySpare 0x6 - Updating 0x7 - Disabled 0x8 - Enabled 0xFF - Unknown
117	Media Type 0x0 - SSD 0x1 - HDD 0xFF - Unknown
118	Protocol 0x0 - Unknown 0x1 - SCSI 0x2 - SAS 0x3 - SATA 0x4 - Fiber 0x5 - VMEE 0x6 - AHCI 0xFF - Other
119	Temperature
120	Slot Number
121	Capable Speed Gbs
122	Negotiated Speed Gbs
123	Progress
124	Belong which controller

	byte	data field
	125	Index in the web page

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x1 0x4
 00 57 44 43 20 57 44 35 30 30 30 42 50 4b 54 2d
 32 32 50 4b 34 54 30 00 00 00 00 00 00 00 00 00
 00 57 44 2d 57 58 4c 31 45 39 31 5a 59 48 56 36
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 31 41 30 31 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 59 65 73 00 00 00 00 00 00 60 c0 70 74 00 00
 00 20 1c 01 01 03 19 04 03 03 00 01 04
```

7.40 Make Dedicated Hotspare for specific volume

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Make Dedicated Hotspare for specific volume = 02h
	5	Drive ID = 00h 0 base
	6:37	Volume Id
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x2 0x7 0x42 0x72 0x6f 0x61 0x64
0x63 0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.41 Make Dedicated Hotspare for specific volume (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Make Dedicated Hotspare for specific volume (Read) = 03h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x3
 00
```

7.42 Make Global Hotspare for all system

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h

	byte	data field
		01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Make Global Hotspare for all system = 04h
	5	Drive ID = 00h 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x4 0x7
00
```

7.43 Make Global Hotspare for all system (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Make Global Hotspare for all system (Read) = 05h

	byte	data field
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x5
 00
```

7.44 Remove Hotspare

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 44h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Remove Hotspare = 06h
	5	Drive ID = 00h 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x6 0x7
 00
```

7.45 Remove Hotspare (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 44h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Remove Hotspare (Read) = 07h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x7
 00
```

7.46 Replace a Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage

	byte	data field
		03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Replace a Drive = 08h
	5	Drive ID 0 base
	6	Drive ID within a Volume 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x8 0x4 0x7
00
```

7.47 Replace a Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Replace a Drive (Read) = 09h
Response Data		

	byte	data field
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0x9
 00
```

7.48 Stop to Replace a Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop to Replace a Drive = 0Ah
	5	Drive ID 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0xA 0x7
 00
```

7.49 Stop to Replace a Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop to Replace a Drive (Read) = 0Bh
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0xB
 00
```

7.50 Indicate LED of specific Physical Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection

	byte	data field
		02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Indicate LED of specific Physical Drive = 0Ch
	5	Drive ID 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0xC 0x7
 00
```

7.51 Indicate LED of specific Physical Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Indicate LED of specific Physical Drive (Read) = 0Dh
Response Data		

	byte	data field
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0xD
 00
```

7.52 Stop to indicate LED of specific Physical Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop to indicate LED of specific Physical Drive = 0Eh
	5	Drive ID 0 base
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0xE 0x7
 00
```

7.53 Stop to indicate LED of specific Physical Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 05h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop to indicate LED of specific Physical Drive (Read) = 0Fh
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x5 0xF
 00
```

7.54 Get Volume Collection Info

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 06h

	byte	data field
		01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Volume Collection Info = 00h
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x6 0x0
 00
```

7.55 Get Volume Collection Info (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 06h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Volume Collection Info (Read) = 01h
Response Data		
	1	Completion Code

	byte	data field
		00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2	Count(s)
	3:~	Volume Name(s)

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x6 0x1
 00 02 42 72 6f 61 64 63 6f 6d 31 30 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 42 72 6f 61 64 63 6f 6d 31 31 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
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 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00
```

7.56 Get Volume Info

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)

	byte	data field
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Get Volume Info = 00h
	5:36	Volume Name
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x0 0x42 0x72 0x6f 0x61 0x64 0x63
0x6f 0x6d 0x30 0x5f 0x31 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.57 Get Volume Info (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume

	byte	data field
	4	Get Volume Info (Read) = 01h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters
	2:33	Name
	34:41	Capacity
	42	Raid Level
	43	Strip Size 0x0 - 16 0x1 - 32 0x2 - 64 0x3 - 128 0x4 - 256 0x5 - 512 0x6 - 1024
	44	State 0x0 - OFFLINE 0x1 - PARTIALLY DEGRADED 0x2 - DEGRADED 0x3 - OPTIMAL
	45	Read Policy 0x0 - Read Ahead 0x1 - No Read Ahead 0xFF - Unknown
	46	Write Policy 0x0 - Write Through 0x1 - Write Back

byte	data field 0x2 - Write Cached
47	0xFF - Unknown IO Policy 0x0 - Direct IO 0x1 - Cached IO 0xFF - Unknown
48	Access Policy 0x0 - Read Write 0x1 - Read Only 0x2 - Blocked 0xFF - Unknown
49	Disk Cache Policy 0x0 - Unchanged 0x1 - Enable 0x2 - Disable 0xFF - Unknown
50	Progress
51	Physical Drive Count
52-179	Physical Drive Members

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x1
 00 54 65 73 74 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 a0 e8 00 00 00 06 03 03 00 00 00 00
 00 00 04 00 01 02 03 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 00 00 00
```

7.58 Check consistency for a specific volume

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Check consistency for a specific volume = 02h
	5:36	Volume ID
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x2 0x42 0x72 0x6f 0x61 0x64 0x63
0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.59 Check consistency for a specific volume (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event

	byte	data field
		05h - Drive 06h - VolumeCollection 07h - Volume
	4	Check consistency for a specific volume (Read) = 03h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x3
 00
```

7.60 Cancel consistency check

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Cancel consistency check = 04h
	5:36	Volume ID
Response Data		

	byte	data field
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x4 0x42 0x72 0x6f 0x61 0x64 0x63
0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.61 Cancel consistency check (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Cancel consistency check (Read) = 05h
	Response Data	
1		Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x5
 00
```

7.62 Indicate LED of specific Logical Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Indicate LED of specific Logical Drive = 06h
	5:36	Volume ID
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x6 0x42 0x72 0x6f 0x61 0x64 0x63
0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.63 Indicate LED of specific Logical Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event

	byte	data field
		05h - Drive 06h - VolumeCollection 07h - Volume
	4	Indicate LED of specific Logical Drive (Read) = 07h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x7
 00
```

7.64 Stop to indicate LED of specific Logical Drive

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop to indicate LED of specific Logical Drive = 08h
	5:36	Volume ID

	byte	data field
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x8 0x42 0x72 0x6f 0x61 0x64 0x63
0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 00
```

7.65 Stop to indicate LED of specific Logical Drive (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Stop to indicate LED of specific Logical Drive (Read) = 09h
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0x9
 00
```

7.66 Change a Virtual Drive Configuration

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Change a Virtual Drive Configuration = 0Ah
	5	Raid Level 0 1 5 6 10 50 60
	6	Number of Drive(s)
	7:38	Drive Ids
	39:70	Volume Name
Response Data		
	1	Completion Code

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0xA 0x6 0x1 0x7 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0x42 0x72 0x6f 0x61 0x64 0x63 0x6f 0x6d 0x31 0x30 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
 00
```

7.67 Change a Virtual Drive Configuration (Read)

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h 01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Change a Virtual Drive Configuration (Read) = 0Bh
Response Data		
	1	Completion Code 00h - success 01h - running 02h - failed 03h - not support 04h - wrong parameters

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0xB
 00
```

7.68 Set Logical Drive Properties

	byte	data field
Request Data	1	Insyde NetFn = Application (06h)
	2	Insyde Sub-Command ID = 47h
	3	Type Selector = 07h

	byte	data field
		01h - StorageCollection 02h - Storage 03h - Controller 04h - Event 05h - Drive 06h - VolumeCollection 07h - Volume
	4	Set Logical Drive Properties = 0Ch
	5:36	Volume Name
	37	Read Policy 0x0 - Read Ahead 0x1 - No Read Ahead
	38	Write Policy 0x0 - Write Through 0x1 - Write Back 0x2 - Write Cached
	39	IO Policy 0x0 - Direct IO 0x1 - Cached IO
	40	Access Policy 0x0 - Read Write 0x1 - Read Only 0x2 - Blocked
	41	Disk Cache Policy 0x0 - Unchanged 0x1 - Enable 0x2 - Disable
	42:73	Rename
Response Data		
	1	Completion Code

	byte	data field

```
spv_ipmi -H $BMCIP -U root -P 0penBmc -I lanplus -C 17 raw 0x3E 0xC0 0x06 0x47 0x7 0xC 0x42 0x72 0x6f 0x61 0x64 0x63
0x6f 0x6d 0x31 0x30 0 0 0
```

8 SNMP

8.1 SNMP Access Control

SNMP access control need to be enabled before acquiring SNMP MIBs resource data.

8.1.1 SNMPv1/v2c Community

NetFn = 0x3E CMD = 0xC0

	Byte	Data Field
Req	1	OEM NetFn: 0Ch
	2	OEM Command ID: 79h OEM command ID for SNMP community string configuration
	3	v1/v2c Set Type 00h = Read Only 01h = Read Write 02h = Enable
	4	v1/v2c Set Flag If Byte 3 Type = 00h/01h 00h = set 01h = disable If Byte 3 Type = 02h 00h = disable 01h = enable
	5:36	Community String (32 bytes)
Rsp	1	Completion Code 00h: Success 01h: Parameter not supported C7h: Request data length invalid CCh: Invalid data field in Request D3h: Destination unavailable

```
1. Set read-only Community "public"
$ ipmitool -I lanplus -U root -P 0penBmc -C 17 -H $BMCIP raw 0x3e 0xc0 0x0c 0x79 0x00 0x00 0x70 0x75 0x62 0x6c 0x69
0x63 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2. Set read-write Community "private"
$ ipmitool -I lanplus -U root -P 0penBmc -C 17 -H $BMCIP raw 0x3e 0xc0 0x0c 0x79 0x01 0x00 0x70 0x72 0x69 0x76 0x61
0x74 0x65 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3. Enable Community
$ ipmitool -I lanplus -U root -P 0penBmc -C 17 -H $BMCIP raw 0x3e 0xc0 0x0c 0x79 0x02 0x01 0x70 0x75 0x62 0x6c 0x69
```

```
0x63 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4. Restart snmpd
$ ipmitool -I lanplus -U root -P 0penBmc -C 17 -H $BMCIP raw 0x3e 0xc0 0x0c 0x01 0x02 0x02 0x00
```

8.1.2 SNMPv3 User

NetFn = 0x3E CMD = 0xC0

	Byte	Data Field
Req	1	OEM NetFn: 0Ch OEM NetFn
	2	OEM Command ID: 71h OEM command ID for SNMP Server v3 user configuration
	3	Action 01h = Add 02h = Delete 03h = Set
	4:19	SNMP v3 User name (16 bytes)
	20	v3 User info (Required when Action Add and Set) [0:1] = SNMP v3 user privacy protocol 00h = No privacy 01h = AES 02h = DES(optional, default is disabled by menuconfig) [2:4] = SNMP v3 user authentication protocol 001b = MD5(1) 010b = SHA1(2) 011b = SHA224(3) 100b = SHA256(4) 101b = SHA384(5) 110b = SHA512(6) [5:6] = SNMP v3 user privilege level 01h = Read Only 02h = Read Write [7]-1b = SNMP v3 user access 0b = disable 1b = enable
	21:32	SNMP v3 user authentication passphrase (Required only when Action Add)
	33:44	SNMP v3 user privacy passphrase (Required only when Action Add and v3 User info privacy protocol AES or DES)
Rsp	1	Completion Code 00h: Success C0h: SNMP daemon is disabled or failed to terminate SNMP daemon. C7h: Request data length invalid CBh: User can't find in IPMI Users list CCh: Invalid data field in Request CDh: 1. At least IPMI User 'User' privilege is required to create SNMP user.

		2. At least IPMI User 'Operator' privilege is required for SNMP users to have Read/Write privilege. D5h: Channel of 802.3 LAN cannot be found.

Add user name "root", passphrase SHA "test123456" and AES "test123456"
$ ipmitool -I lanplus -U root -P 0penBmc -C 17 -H $BMCIP raw 0x3e 0xc0 0x0c 0x71 0x01 0x72 0x6f 0x6f 0x74 0 0 0 0 0 0 0 0 0 0 0 0 0xc9 0x74 0x65 0x73 0x74 0x31 0x32 0x33 0x34 0x35 0x36 0 0 0x74 0x65 0x73 0x74 0x31 0x32 0x33 0x34 0x35 0x36 0 0

8.2 SNMP MIBs for RAID

8.2.1 OID detail

1.3.6.1.4.1.45065.1.7.1.0 - storageRefresh
1.3.6.1.4.1.45065.1.7.2.0 – storageRaidCount

OID Name	Type	Access	Description
storageRefresh	Integer(1)	W	Set refresh storage information(1)(e.g. set 1 to refresh storage)
storageRaidCount	String	R	Get RAID count(e.g. Raid:1,drive:1,controller:1,volume:1)

*Table Column OID RAID drive

index: drive count index

1.3.6.1.4.1.45065.1.7.3.1.1.(index) - srdDesc
1.3.6.1.4.1.45065.1.7.3.1.2.(index) - srdName
1.3.6.1.4.1.45065.1.7.3.1.3.(index) - srdStatusIndicator
1.3.6.1.4.1.45065.1.7.3.1.4.(index) - srdIndicatorLED
1.3.6.1.4.1.45065.1.7.3.1.5.(index) - srdModel
1.3.6.1.4.1.45065.1.7.3.1.6.(index) - srdRevision
1.3.6.1.4.1.45065.1.7.3.1.7.(index) - srdState
1.3.6.1.4.1.45065.1.7.3.1.8.(index) - srdHealth
1.3.6.1.4.1.45065.1.7.3.1.9.(index) - srdHealthRollup
1.3.6.1.4.1.45065.1.7.3.1.10.(index) - srdCapacityBytes
1.3.6.1.4.1.45065.1.7.3.1.11.(index) - srdFailurePredicted
1.3.6.1.4.1.45065.1.7.3.1.12.(index) - srdProtocol
1.3.6.1.4.1.45065.1.7.3.1.13.(index) - srdMediaType
1.3.6.1.4.1.45065.1.7.3.1.14.(index) - srdManufacturer
1.3.6.1.4.1.45065.1.7.3.1.15.(index) - srdSKU
1.3.6.1.4.1.45065.1.7.3.1.16.(index) - srdSerialNum
1.3.6.1.4.1.45065.1.7.3.1.17.(index) - srdPartNum
1.3.6.1.4.1.45065.1.7.3.1.18.(index) - srdAssetTag
1.3.6.1.4.1.45065.1.7.3.1.19.(index) - srdHotsparetype
1.3.6.1.4.1.45065.1.7.3.1.20.(index) - srdEncryptionAbility
1.3.6.1.4.1.45065.1.7.3.1.21.(index) - srdEncryptionStatus
1.3.6.1.4.1.45065.1.7.3.1.22.(index) - srdRotationSpeedRPM
1.3.6.1.4.1.45065.1.7.3.1.23.(index) - srdBlockSizeBytes
1.3.6.1.4.1.45065.1.7.3.1.24.(index) - srdCapableSpeedGBS
1.3.6.1.4.1.45065.1.7.3.1.25.(index) - srdNegotiatedSpeedGBS
1.3.6.1.4.1.45065.1.7.3.1.26.(index) - srdPredictedMediaLifeLeftPercent
1.3.6.1.4.1.45065.1.7.3.1.27.(index) - srdOemTemp

1.3.6.1.4.1.45065.1.7.3.1.28.(index) - srdOemSlotNum
1.3.6.1.4.1.45065.1.7.3.1.29.(index) - srdOemPatrolReadProgressPercent
1.3.6.1.4.1.45065.1.7.3.1.30.(index) - srdOemCopyBackProgressPercent
1.3.6.1.4.1.45065.1.7.3.1.31.(index) - srdStorageId
1.3.6.1.4.1.45065.1.7.3.1.32.(index) - srdVolumes
1.3.6.1.4.1.45065.1.7.3.1.33.(index) - srdMakeDedicatedHotSpare
1.3.6.1.4.1.45065.1.7.3.1.34.(index) - srdMakeGlobalHotSpare
1.3.6.1.4.1.45065.1.7.3.1.35.(index) - srdRemoveHotSpare

37. Table Column Definition RAID drive

OID Name	Type	Access	Description
srdDesc	String	R	Storage RAID drive description
srdName	String	R	Storage RAID drive name
srdStatusIndicator	String	R	Storage RAID drive status indicator
srdIndicatorLED	String	R	Storage RAID drive indicator led
srdModel	String	R	Storage RAID drive model
srdRevision	String	R	Storage RAID drive revision
srdState	String	R	Storage RAID drive state
srdHealth	String	R	Storage RAID drive health
srdHealthRollup	String	R	Storage RAID drive health rollup
srdCapacityBytes	String	R	Storage RAID drive capacity bytes
srdFailurePredicted	String	R	Storage RAID drive failure predicted
srdProtocol	String	R	Storage RAID drive protocol
srdMediaType	String	R	Storage RAID drive media type
srdManufacturer	String	R	Storage RAID drive manufacturer
srdSKU	String	R	Storage RAID drive sku
srdSerialNum	String	R	Storage RAID drive serial number
srdPartNum	String	R	Storage RAID drive part number
srdAssetTag	String	R	Storage RAID drive asset tag
srdHotsparetype	String	R/W	Storage RAID drive hotsparetype
srdEncryptionAbility	String	R	Storage RAID drive encryption ability

OID Name	Type	Access	Description
srdEncryptionStatus	String	R	Storage RAID drive encryption status
srdRotationSpeedRPM	String	R	Storage RAID drive rotation speedrpm
srdBlockSizeBytes	String	R	Storage RAID drive block size bytes
srdCapableSpeedGBS	String	R	Storage RAID drive capable speed gbs
srdNegotiatedSpeedGBS	String	R	Storage RAID drive negotiated speed gbs
srdPredictedMediaLifeLeftPercent	String	R	Storage RAID drive predicted media life left percent
srdOemTemp	String	R	Storage RAID drive oem temperature
srdOemSlotNum	String	R	Storage RAID drive oem slot number
srdOemPatrolReadProgressPercent	String	R	Storage RAID drive oem patrol read progress percent
srdOemCopyBackProgressPercent	String	R	Storage RAID drive oem copy back progress percent
srdStorageId	String	R	Storage RAID drive storage id
srdVolumes	String	R	Storage RAID drive created volume id
srdMakeDedicatedHotSpare	Integer32	R/W	Storage RAID drive make dedicated hot spare
srdMakeGlobalHotSpare	Integer32	R/W	Storage RAID drive make global hot spare
srdRemoveHotSpare	Integer32	R/W	Storage RAID drive remove hot spare

*Table Column OID RAID controller

index: controller count index

1.3.6.1.4.1.45065.1.7.4.1.1.(index) - srdDesc
1.3.6.1.4.1.45065.1.7.4.1.2.(index) - srdName
1.3.6.1.4.1.45065.1.7.4.1.3.(index) - srcState
1.3.6.1.4.1.45065.1.7.4.1.4.(index) - srcHealth
1.3.6.1.4.1.45065.1.7.4.1.5.(index) - srcHealthRollup
1.3.6.1.4.1.45065.1.7.4.1.6.(index) - srcSpeedGBPS
1.3.6.1.4.1.45065.1.7.4.1.7.(index) - srcFwVer
1.3.6.1.4.1.45065.1.7.4.1.8.(index) - srcPackageVer
1.3.6.1.4.1.45065.1.7.4.1.9.(index) - srcManufacturer

1.3.6.1.4.1.45065.1.7.4.1.10.(index) - srcModel
1.3.6.1.4.1.45065.1.7.4.1.11.(index) - srcSKU
1.3.6.1.4.1.45065.1.7.4.1.12.(index) - srcSerialNum
1.3.6.1.4.1.45065.1.7.4.1.13.(index) - srcPartNum
1.3.6.1.4.1.45065.1.7.4.1.14.(index) - srcAssetTag
1.3.6.1.4.1.45065.1.7.4.1.15.(index) - srcSupportedControllerProtocols
1.3.6.1.4.1.45065.1.7.4.1.16.(index) - srcSupportedDeviceProtocols
1.3.6.1.4.1.45065.1.7.4.1.17.(index) - srcOemSeqNoNewestEvent
1.3.6.1.4.1.45065.1.7.4.1.18.(index) - srcOemSeqNoOldestEvent
1.3.6.1.4.1.45065.1.7.4.1.19.(index) - srcOemSeqNoLastClearEvent
1.3.6.1.4.1.45065.1.7.4.1.20.(index) - srcOemSeqNoLastCleanShutdownEvent
1.3.6.1.4.1.45065.1.7.4.1.21.(index) - srcOemSeqNoThisSessionBootEvent
1.3.6.1.4.1.45065.1.7.4.1.22.(index) - srcOemBBUStatus
1.3.6.1.4.1.45065.1.7.4.1.23.(index) - srcOemBBUTempCelsius
1.3.6.1.4.1.45065.1.7.4.1.24.(index) - srcOemBBUVoltMillivolts
1.3.6.1.4.1.45065.1.7.4.1.25.(index) - srcOemBBUCurMilliamps
1.3.6.1.4.1.45065.1.7.4.1.26.(index) - srcOemBBUSerialNum
1.3.6.1.4.1.45065.1.7.4.1.27.(index) - srcOemBBUHealthPercent
1.3.6.1.4.1.45065.1.7.4.1.28.(index) - srcOemBBURemainingBatteryPercent
1.3.6.1.4.1.45065.1.7.4.1.29.(index) - srcOemJunctionTemp
1.3.6.1.4.1.45065.1.7.4.1.30.(index) - srcOemMemory
1.3.6.1.4.1.45065.1.7.4.1.31.(index) - srcStorageId

38. Table Column Definition RAID Controller

OID Name	Type	Access	Description
srcDesc	String	R	Storage RAID controller description
srcName	String	R	Storage RAID controller name
srcState	String	R	Storage RAID controller state
srcHealth	String	R	Storage RAID controller health
srcHealthRollup	String	R	Storage RAID controller health rollup
srcSpeedGBPS	String	R	Storage RAID controller speed gbps
srcFwVer	String	R	Storage RAID controller firmware version
srcPackageVer	String	R	Storage RAID controller package version
srcManufacturer	String	R	Storage RAID controller manufacturer
srcModel	String	R	Storage RAID controller model
srcSKU	String	R	Storage RAID controller sku
srcSerialNum	String	R	Storage RAID controller serial number
srcPartNum	String	R	Storage RAID controller part number
srcAssetTag	String	R	Storage RAID controller asset tag
srcSupportedController Protocols	String	R	Storage RAID controller supported controller protocols
srcSupportedDevicePro tocols	String	R	Storage RAID controller supported device protocols

OID Name	Type	Access	Description
srcOemSeqNoNewestEvent	String	R	Storage RAID controller oem seqno newest event
srcOemSeqNoOldestEvent	String	R	Storage RAID controller oem seqno oldest event
srcOemSeqNoLastClearEvent	String	R	Storage RAID controller oem seqno last clear event
srcOemSeqNoLastCleanShutdownEvent	String	R	Storage RAID controller oem seqno last clean shutdown event
srcOemSeqNoThisSessionBootEvent	String	R	Storage RAID controller oem seqno this session boot event
srcOemBBUStatus	String	R	Storage RAID controller oem bbu status
srcOemBBUTempCelsius	String	R	Storage RAID controller oem bbu temp celsius
srcOemBBUVoltMillivolts	String	R	Storage RAID controller oem bbu volt millivolts
srcOemBBUCurMilliamps	String	R	Storage RAID controller oem bbu cur milliamps
srcOemBBUSerialNum	String	R	Storage RAID controller oem bbu serial number
srcOemBBUHealthPercent	String	R	Storage RAID controller oem bbu health percent
srcOemBBURemainingBatteryPercent	String	R	Storage RAID controller oem bbu remaining battery percent
srcOemJunctionTemp	String	R	Storage RAID controller oem junction temp
srcOemMemory	String	R	Storage RAID controller oem memory
srcStorageId	String	R	Storage RAID controller storage id

*Table Column OID RAID volume

index: volume count index

1.3.6.1.4.1.45065.1.7.5.1.1.(index) - srvDesc
1.3.6.1.4.1.45065.1.7.5.1.2.(index) - srvName
1.3.6.1.4.1.45065.1.7.5.1.3.(index) - srvState
1.3.6.1.4.1.45065.1.7.5.1.4.(index) - srvHealth
1.3.6.1.4.1.45065.1.7.5.1.5.(index) - srvHealthRollup
1.3.6.1.4.1.45065.1.7.5.1.6.(index) - srvCapacityBytes
1.3.6.1.4.1.45065.1.7.5.1.7.(index) - srvVolumeType
1.3.6.1.4.1.45065.1.7.5.1.8.(index) - srvEncrypted
1.3.6.1.4.1.45065.1.7.5.1.9.(index) - srvEncryptionType
1.3.6.1.4.1.45065.1.7.5.1.10.(index) - srvBlockSizeBytes
1.3.6.1.4.1.45065.1.7.5.1.11.(index) - srvOperations
1.3.6.1.4.1.45065.1.7.5.1.12.(index) - srvOptimumIosizeBytes
1.3.6.1.4.1.45065.1.7.5.1.13.(index) - srvOemRaidLevel
1.3.6.1.4.1.45065.1.7.5.1.14.(index) - srvOemState

1.3.6.1.4.1.45065.1.7.5.1.15.(index) - srvOemReadPolicy
1.3.6.1.4.1.45065.1.7.5.1.16.(index) - srvOemWritePolicy
1.3.6.1.4.1.45065.1.7.5.1.17.(index) - srvOemIOPolicy
1.3.6.1.4.1.45065.1.7.5.1.18.(index) - srvOemAccessPolicy
1.3.6.1.4.1.45065.1.7.5.1.19.(index) - srvOemDiskCachePolicy
1.3.6.1.4.1.45065.1.7.5.1.20.(index) - srvOemStripSize
1.3.6.1.4.1.45065.1.7.5.1.21.(index) - srvOemBadBlocks
1.3.6.1.4.1.45065.1.7.5.1.22.(index) - srvStorageId
1.3.6.1.4.1.45065.1.7.5.1.23.(index) - srvDrives
1.3.6.1.4.1.45065.1.7.5.1.24.(index) - srvFuncDelete
1.3.6.1.4.1.45065.1.7.5.1.25.(index) - srvPatchAction

39. Table Column Definition RAID volume

OID Name	Type	Access	Description
srvDesc	String	R	Storage RAID volume description
srvName	String	R/W	Storage RAID volume name
srvState	String	R	Storage RAID volume state
srvHealth	String	R	Storage RAID volume health
srvHealthRollup	String	R	Storage RAID volume health rollup
srvCapacityBytes	String	R	Storage RAID volume capacity bytes
srvVolumeType	String	R	Storage RAID volume volume type
srvEncrypted	String	R	Storage RAID volume encrypted
srvEncryptionType	String	R	Storage RAID volume encryption type
srvBlockSizeBytes	String	R	Storage RAID volume block size bytes
srvOperations	String	R	Storage RAID volume operations
srvOptimumIosizeBytes	String	R	Storage RAID volume optimum iosize bytes
srvOemRaidLevel	String	R	Storage RAID volume oem raid level
srvOemState	String	R	Storage RAID volume oem state
srvOemReadPolicy	String	R/W	Storage RAID volume oem read policy
srvOemWritePolicy	String	R/W	Storage RAID volume oem write policy
srvOemIOPolicy	String	R/W	Storage RAID volume oem io policy
srvOemAccessPolicy	String	R/W	Storage RAID volume oem access policy
srvOemDiskCachePolicy	String	R/W	Storage RAID volume oem disk cache policy
srvOemStripSize	String	R	Storage RAID volume oem strip size
srvOemBadBlocks	String	R	Storage RAID volume oem bad blocks
srvStorageId	String	R	Storage RAID volume storage id

OID Name	Type	Access	Description
srvDrives	String	R	Storage RAID volume drives link
srvFuncDelete	Integer	R/W	Storage RAID volume function delete
srvPatchAction	Integer	R/W	Storage RAID volume patch action

*RAID event log

1.3.6.1.4.1.45065.1.7.6.0 - storageRaidEventController
1.3.6.1.4.1.45065.1.7.7.0 - storageRaidEventCount
1.3.6.1.4.1.45065.1.7.8.0 - storageRaidEventStartIndex

OID Name	Type	Access	Description
storageRaidEventController	String	Read-Write	Storage RAID event controller(e.g. storage id,controller id: 1,1)
storageRaidEventCount	Integer	Read-Only	Get Storage RAID event count
storageRaidEventStartIndex	Integer	Read-Write	Storage RAID event start index

*Table Column OID RAID event

index: 1~35

1.3.6.1.4.1.45065.1.7.9.1.1.(index) - sreSeqno
1.3.6.1.4.1.45065.1.7.9.1.2.(index) - sreLevel
1.3.6.1.4.1.45065.1.7.9.1.3.(index) - sreTime
1.3.6.1.4.1.45065.1.7.9.1.4.(index) - sreDesc

Table Column Definition RAID event

OID Name	Type	Access	Description
sreSeqno	Integer	R	Storage RAID event log sequence number
sreLevel	String	R	Storage RAID event log level
sreTime	String	R	Storage RAID event log time
sreDesc	String	R	Storage RAID event log description

*Table Column OID RAID controller Function

index: controller count index

1.3.6.1.4.1.45065.1.7.12.1.1.(index) - srcfDeleteVolume
1.3.6.1.4.1.45065.1.7.12.1.2.(index) - srcfGetFreeRaidResource
1.3.6.1.4.1.45065.1.7.12.1.3.(index) - srcfCreateVolumeBasicData
1.3.6.1.4.1.45065.1.7.12.1.4.(index) - srcfCreateVolumeParam
1.3.6.1.4.1.45065.1.7.12.1.5.(index) - srcfSaveTTYLog
1.3.6.1.4.1.45065.1.7.12.1.6.(index) - srcfScanForeignConfig
1.3.6.1.4.1.45065.1.7.12.1.7.(index) - srcfImportForeignConfig
1.3.6.1.4.1.45065.1.7.12.1.8.(index) - srcfClearForeignConfig
1.3.6.1.4.1.45065.1.7.12.1.9.(index) - srcfStartPatrolRead
1.3.6.1.4.1.45065.1.7.12.1.10.(index) - srcfStopPatrolRead

Table Column Definition RAID Controller Function

OID Name	Type	Access	Description
srcfDeleteVolume	Integer	R/W	Storage RAID controller function delete volume
srcfGetFreeRaidResource	Integer	R/W	Storage RAID controller function get free raid resource
srcfCreateVolumeBasicData	Integer	R/W	Storage RAID controller function create volume basic data
srcfCreateVolumeParam	Integer	R/W	Storage RAID controller function create volume parameter
srcfSaveTTYLog	Integer	R/W	Storage RAID controller function save tty log
srcfScanForeignConfig	Integer	R/W	Storage RAID controller function scan foreign configuration
srcfImportForeignConfig	Integer	R/W	Storage RAID controller function import foreign configuration
srcfClearForeignConfig	Integer	R/W	Storage RAID controller function clear foreign configuration
srcfStartPatrolRead	Integer	R/W	Storage RAID controller function start patrol read
srcfStopPatrolRead	Integer	R/W	Storage RAID controller function stop patrol read

*Table Column OID RAID controller Function Create Volume Basic Data Raid Level

index: RAID level count index

1.3.6.1.4.1.45065.1.7.13.1.1.(index) - srcfCvbdRLRaidLevel

40. Table Column Definition Raid Controller Function

OID Name	Type	Access	Description
srcfCvbdRLRaidLevel	String	R/W	Storage RAID controller function create volume basic data raid level parameter

*Table Column OID RAID controller Function Create Volume Basic Data Free Drive Id

index: free drive id count index

1.3.6.1.4.1.45065.1.7.14.1.1.(index) - srcfCvbdFDIFreeDriveId

41. Table Column Definition RAID Controller Function Create Volume Basic Data Free Drive Id

OID Name	Type	Access	Description
srcfCvbdFDIFreeDriveId	String	R/W	Storage RAID controller function create volume basic data free drive id parameter

*OID RAID controller Function Create Volume Parameter Name and Size

1.3.6.1.4.1.45065.1.7.15.1.0 - srcfCVPName
1.3.6.1.4.1.45065.1.7.16.1.0 - srcfCVPSize

42. Definition RAID Controller Function Create Volume Parameter Name

OID Name	Type	Access	Description
srcfCVPName	String	R/W	Storage RAID controller function create volume parameter name
srcfCVPSize	Integer	R/W	Storage RAID controller function create volume parameter size

*Table Column OID RAID controller Function Create Volume Parameter Stripe Size

index: stripe size count index

1.3.6.1.4.1.45065.1.7.17.1.1.(index) - srcfCvpSSStripeSize

43. Table Column Definition RAID Controller Function Create Volume Parameter Stripe Size

OID Name	Type	Access	Description
srcfCvpSSStripeSize	Integer	R/W	Storage RAID controller function create volume parameter stripe size parameter

*Table Column OID RAID controller Function Create Volume Parameter Access Policy

index: access policy count index

1.3.6.1.4.1.45065.1.7.18.1.1.(index) - srcfCvpAPAccessPolicy

44. Table Column Definition RAID Controller Function Create Volume Parameter Access Policy

OID Name	Type	Access	Description
srcfCvpAPAccessPolicy	String	R/W	Storage RAID controller function create volume parameter access policy parameter

*Table Column OID RAID controller Function Create Volume Parameter Disk Cache

index: disk cache count index

1.3.6.1.4.1.45065.1.7.19.1.1.(index) - srcfCvpDCPDiskCachePolicy

45. Table Column Definition RAID Controller Function Create Volume Parameter Disk Cache

OID Name	Type	Access	Description
srcfCvpDCPDiskCachePolicy	String	R/W	Storage RAID controller function create volume parameter disk cache policy parameter

*Table Column OID RAID controller Function Create Volume Parameter Init State

index: init state count index

1.3.6.1.4.1.45065.1.7.20.1.1.(index) - srcfCvpISInitState

46. Table Column Definition RAID Controller Function Create Volume Parameter Init State

OID Name	Type	Access	Description
srcfCvpISInitState	String	R/W	Storage RAID controller function create volume parameter init state parameter

*OID RAID controller Function Function TTY Log

```
1.3.6.1.4.1.45065.1.7.21.1.0 - srcfTLStartIndex
1.3.6.1.4.1.45065.1.7.22.1.0 - srcfTLCount
```

47. Definition RAID Controller Function Function TTY Log

OID Name	Type	Access	Description
srcfTLStartIndex	Integer	R/W	The start index of Storage RAID controller function tty log
srcfTLCount	Integer	R/W	The total count of Storage RAID controller function tty log

*Table Column OID RAID controller Function TTY Log

```
index: tty log count index

1.3.6.1.4.1.45065.1.7.23.1.1.(index) - srcfTLTTYLog
```

48. Table Column Definition RAID Controller Function TTY Log

OID Name	Type	Access	Description
srcfTLTTYLog	String	R	Storage RAID controller function tty log

*Table Column OID RAID controller Function Scan Foreign Configuration

```
1.3.6.1.4.1.45065.1.7.24.1.1.0 - srcfSFCTargetId
1.3.6.1.4.1.45065.1.7.24.1.2.0 - srcfSFCRaidLevel
1.3.6.1.4.1.45065.1.7.24.1.3.0 - srcfSFCSize
1.3.6.1.4.1.45065.1.7.24.1.4.0 - srcfSFCState
```

49. Table Column Definition RAID Controller Function Scan Foreign Configuration

OID Name	Type	Access	Description
srcfSFCTargetId	String	R	Storage RAID controller function scan foreign configuration target id
srcfSFCRaidLevel	String	R	Storage RAID controller function scan foreign configuration raid level
srcfSFCSize	Integer	R	Storage RAID controller function scan foreign configuration size
srcfSFCState	String	R	Storage RAID controller function scan foreign configuration state

8.2.2 Command usage

SNMP RAID Only available when host is power on Set storageRefresh - Set refresh storage information including raid, event log and sata

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.1.0 i 1
```

If SNMP RAID refresh command fail, please check service **xyz.openbmc_project.State.Host** CurrentHostState.

```
busctl get-property xyz.openbmc_project.State.Host /xyz/openbmc_project/state/host0 xyz.openbmc_project.State.Host CurrentHostState
```

When CurrentHostState is **xyz.openbmc_project.State.Host.HostState.On**, SNMP RAID refresh will allow you to do further action.

8.2.2.1 RAID information

Get storageRaidCount - Get RAID count(e.g. Raid:1,drive:1,controller:1,volume:1)

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.2.0
SNMPv2-SMI::enterprises.45065.1.7.2.0 = STRING: "Raid:1,drive:4,controller:1,volume:1"
```

Get RAID drive table

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.3
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.1.1 = STRING: "Physical Disk 1"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.2.1 = STRING: "Physical Disk 1"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.3.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.4.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.5.1 = STRING: "Hitachi HTS72505"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.6.1 = STRING: "C70E"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.7.1 = STRING: "StandbyOffline"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.8.1 = STRING: "OK"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.9.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.10.1 = STRING: "500107862016"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.12.1 = STRING: "SATA"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.13.1 = STRING: "HDD"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.14.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.15.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.16.1 = STRING: "110211PCK400GLGTB09J"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.17.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.18.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.19.1 = STRING: "None"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.20.1 = STRING: "None"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.21.1 = STRING: "Unencrypted"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.23.1 = STRING: "512"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.24.1 = STRING: "3.0"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.25.1 = STRING: "3.0"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.27.1 = STRING: "28"
...
```

```
SNMPv2-SMI::enterprises.45065.1.7.3.1.28.1 = STRING: "1"
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.29.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.30.1 = ""
...
SNMPv2-SMI::enterprises.45065.1.7.3.1.31.1 = STRING: "1"
...
```

Get RAID controller table

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.4
SNMPv2-SMI::enterprises.45065.1.7.4.1.1.0 = STRING: "Broadcom RAID Controller"
SNMPv2-SMI::enterprises.45065.1.7.4.1.2.0 = STRING: "Broadcom"
SNMPv2-SMI::enterprises.45065.1.7.4.1.3.0 = STRING: "Enabled"
SNMPv2-SMI::enterprises.45065.1.7.4.1.4.0 = STRING: "OK"
SNMPv2-SMI::enterprises.45065.1.7.4.1.5.0 = STRING: "OK"
SNMPv2-SMI::enterprises.45065.1.7.4.1.6.0 = STRING: "12.000000"
SNMPv2-SMI::enterprises.45065.1.7.4.1.7.0 = STRING: "5.010.00-0671"
SNMPv2-SMI::enterprises.45065.1.7.4.1.8.0 = STRING: "50.1.0-0501"
SNMPv2-SMI::enterprises.45065.1.7.4.1.9.0 = STRING: "Broadcom"
SNMPv2-SMI::enterprises.45065.1.7.4.1.10.0 = STRING: "AVAGO MegaRAID SAS 9460-16i"
SNMPv2-SMI::enterprises.45065.1.7.4.1.11.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.12.0 = STRING: "SP74516538"
SNMPv2-SMI::enterprises.45065.1.7.4.1.13.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.14.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.15.0 = STRING: "PCIe"
SNMPv2-SMI::enterprises.45065.1.7.4.1.16.0 = STRING: "SAS,SATA"
SNMPv2-SMI::enterprises.45065.1.7.4.1.17.0 = STRING: "15290"
SNMPv2-SMI::enterprises.45065.1.7.4.1.18.0 = STRING: "1"
SNMPv2-SMI::enterprises.45065.1.7.4.1.19.0 = STRING: "219"
SNMPv2-SMI::enterprises.45065.1.7.4.1.20.0 = STRING: "14596"
SNMPv2-SMI::enterprises.45065.1.7.4.1.21.0 = STRING: "15279"
SNMPv2-SMI::enterprises.45065.1.7.4.1.22.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.23.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.24.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.25.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.26.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.27.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.28.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.4.1.29.0 = STRING: "0"
SNMPv2-SMI::enterprises.45065.1.7.4.1.30.0 = STRING: "4096"
SNMPv2-SMI::enterprises.45065.1.7.4.1.31.0 = STRING: "1"
```

Get RAID volume table

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.5
SNMPv2-SMI::enterprises.45065.1.7.5.1.1.0 = STRING: "Logical Disk Broadcom10"
SNMPv2-SMI::enterprises.45065.1.7.5.1.2.0 = STRING: "EikTest"
SNMPv2-SMI::enterprises.45065.1.7.5.1.3.0 = STRING: "Enabled"
SNMPv2-SMI::enterprises.45065.1.7.5.1.4.0 = STRING: "OK"
SNMPv2-SMI::enterprises.45065.1.7.5.1.5.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.5.1.6.0 = STRING: "1499212021760"
SNMPv2-SMI::enterprises.45065.1.7.5.1.7.0 = STRING: "NonRedundant"
SNMPv2-SMI::enterprises.45065.1.7.5.1.8.0 = STRING: "0"
SNMPv2-SMI::enterprises.45065.1.7.5.1.9.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.5.1.10.0 = STRING: "512"
SNMPv2-SMI::enterprises.45065.1.7.5.1.11.0 = ""
SNMPv2-SMI::enterprises.45065.1.7.5.1.12.0 = STRING: "512"
SNMPv2-SMI::enterprises.45065.1.7.5.1.13.0 = STRING: "0"
SNMPv2-SMI::enterprises.45065.1.7.5.1.14.0 = STRING: "OPTIMAL"
SNMPv2-SMI::enterprises.45065.1.7.5.1.15.0 = STRING: "Read Ahead"
SNMPv2-SMI::enterprises.45065.1.7.5.1.16.0 = STRING: "Write Through"
```

```
SNMPv2-SMI::enterprises.45065.1.7.5.1.17.0 = STRING: "Direct IO"
SNMPv2-SMI::enterprises.45065.1.7.5.1.18.0 = STRING: "Read Write"
SNMPv2-SMI::enterprises.45065.1.7.5.1.19.0 = STRING: "Unchanged"
SNMPv2-SMI::enterprises.45065.1.7.5.1.20.0 = STRING: "64"
SNMPv2-SMI::enterprises.45065.1.7.5.1.21.0 = STRING: "0"
SNMPv2-SMI::enterprises.45065.1.7.5.1.22.0 = STRING: "1"
```

8.2.2.2 Hot Spare

Requirement

1. One RAID volume established e.g. RAID1
2. One empty hard disk at least

Dedicated Set hotspare
Drive Hot Spare Type OID - 1.7.3.1.19.(drive id)

Set with **volume id**

```
E.g. RAID1 created with drive0 and drive1, volume id = 0.

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.19.2 s "0"
```

Set makeDedicatedHotSpare
Must have **volume id** set first

Make Dedicated Hot Spare OID - 1.7.3.1.33.(drive id)

```
E.g. There are 3 drives, id: 0, 1, 2. RAID1 volume created with drive0, drive1. Drive2 is empty.

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.33.2 i 1
```

If success, selected drive hot spare type will change to dedicated.

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.19.2
SNMPv2-SMI::enterprises.45065.1.7.3.1.19.2 = STRING: "Dedicated"
```

Global Set makeGlobalHotSpare
Make Global Hot Spare OID - 1.7.3.1.34.(drive id)

```
E.g. There are 3 drives, id: 0, 1, 2. Raid1 volume created with drive0, drive1. Drive2 is empty.

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.34.2 i 1
```

If success, selected drive hot spare type will change to global.

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.19.2
SNMPv2-SMI::enterprises.45065.1.7.3.1.19.2 = STRING: "Global"
```

Remove Set removeHotSpare
Remove Hot Spare OID - 1.7.3.1.35.(drive id)

```
E.g. Drive2 is set with hot spare dedicated or global.

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.35.2 i 1
```

If success, selected drive hot spare type will be removed.

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.3.1.19.2
SNMPv2-SMI::enterprises.45065.1.7.3.1.19.2 = STRING: "None"
```

8.2.2.3 Foreign configuration

50. **Scan Foreign Configuration**

Set srcfScanForeignConfig to start the scanning

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.6.0 i 1
```

51. **Get Foreign Configuration**

```
1.3.6.1.4.1.45065.1.7.24.1.1.0 - srcfSFCTargetId
$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.1.0
iso.3.6.1.4.1.45065.1.7.24.1.1.0 = STRING: "0"

1.3.6.1.4.1.45065.1.7.24.1.2.0 - srcfSFCRaidLevel

$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.2.0
iso.3.6.1.4.1.45065.1.7.24.1.2.0 = STRING: "1"

1.3.6.1.4.1.45065.1.7.24.1.3.0 - srcfSFCSize

$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.3.0
iso.3.6.1.4.1.45065.1.7.24.1.3.0 = INTEGER: 476416

1.3.6.1.4.1.45065.1.7.24.1.4.0 - srcfSFCState

$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.4.0
iso.3.6.1.4.1.45065.1.7.24.1.4.0 = STRING: "OPTIMAL"
```

52. **Import Foreign Configuration**

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.7.0 i 1
```

If foreign configuration import success, the same volume setup will be generated automatically check result by refreshStorage and then get volume table

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.1.0 i
1
iso.3.6.1.4.1.45065.1.7.1.0 = INTEGER: 1
$ snmpwalk -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.5
iso.3.6.1.4.1.45065.1.7.5.1.1.0 = STRING: "Logical Disk Broadcom0_1"
iso.3.6.1.4.1.45065.1.7.5.1.2.0 = STRING: "TestP"
iso.3.6.1.4.1.45065.1.7.5.1.3.0 = STRING: "Enabled"
iso.3.6.1.4.1.45065.1.7.5.1.4.0 = STRING: "OK"
iso.3.6.1.4.1.45065.1.7.5.1.5.0 = ""
iso.3.6.1.4.1.45065.1.7.5.1.6.0 = STRING: "499558383616"
iso.3.6.1.4.1.45065.1.7.5.1.7.0 = STRING: "Mirrored"
iso.3.6.1.4.1.45065.1.7.5.1.8.0 = STRING: "0"
iso.3.6.1.4.1.45065.1.7.5.1.9.0 = ""
iso.3.6.1.4.1.45065.1.7.5.1.10.0 = STRING: "512"
iso.3.6.1.4.1.45065.1.7.5.1.11.0 = ""
iso.3.6.1.4.1.45065.1.7.5.1.12.0 = STRING: "512"
```

```
iso.3.6.1.4.1.45065.1.7.5.1.13.0 = STRING: "1"
iso.3.6.1.4.1.45065.1.7.5.1.14.0 = STRING: "Optimal"
iso.3.6.1.4.1.45065.1.7.5.1.15.0 = STRING: "No Read Ahead"
iso.3.6.1.4.1.45065.1.7.5.1.16.0 = STRING: "Write Back"
iso.3.6.1.4.1.45065.1.7.5.1.17.0 = STRING: "Direct IO"
iso.3.6.1.4.1.45065.1.7.5.1.18.0 = STRING: "Read Write"
iso.3.6.1.4.1.45065.1.7.5.1.19.0 = STRING: "Disable"
iso.3.6.1.4.1.45065.1.7.5.1.20.0 = STRING: "64"
iso.3.6.1.4.1.45065.1.7.5.1.21.0 = STRING: "0"
iso.3.6.1.4.1.45065.1.7.5.1.22.0 = STRING: "1"
iso.3.6.1.4.1.45065.1.7.5.1.23.0 = STRING: "0,1"
iso.3.6.1.4.1.45065.1.7.5.1.24.0 = INTEGER: 0
iso.3.6.1.4.1.45065.1.7.5.1.25.0 = INTEGER: 0
iso.3.6.1.4.1.45065.1.7.5.1.26.0 = INTEGER: 0
iso.3.6.1.4.1.45065.1.7.5.1.27.0 = INTEGER: 0
```

53. **Clear Foreign Configuration**

First, scan foreign configuration. Make sure the foreign configuration exist

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.6.0 i 1
$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.1.0
iso.3.6.1.4.1.45065.1.7.24.1.1.0 = STRING: "0"
$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.2.0
iso.3.6.1.4.1.45065.1.7.24.1.2.0 = STRING: "1"
$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.3.0
iso.3.6.1.4.1.45065.1.7.24.1.3.0 = INTEGER: 476416
$ snmpget -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.24.1.4.0
iso.3.6.1.4.1.45065.1.7.24.1.4.0 = STRING: "OPTIMAL"
```

Then delete foreign configuration

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.8.0 i 1
```

Then, scan again. There should be no foreign configuration properties

54. **Start Patrol read**

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.9.0 i 1
```

55. **Stop Patrol read**

```
$ snmpset -v 3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.10.0 i 1
```

8.2.2.4 Create volume

1. **Set getFreeRaidResource**
Controller Function getFreeRaidResource OID - 1.7.12.1.2.(controller id)

```
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.2.0 i 1
```

If **success**, there will be **raid level** and **free drive id** data available.
2. **RAID level**
:**Get RAID level**

RAID Level OID - 1.7.13.1.1.(available RAID level count)

```
E.g. There are raid0 and raid1 available.

$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.13
SNMPv2-SMI::enterprises.45065.1.7.13.1.1.0 = STRING: "0"
SNMPv2-SMI::enterprises.45065.1.7.13.1.1.1 = STRING: "1"
```

Set RAID level
Can only select one RAID level each time

```
E.g. Select RAID1

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.13.1.1.1 s "1"
```

3. **Free drive id**
:**Get free drive id**
Free Drive Id OID - 1.7.14.1.1.(available free drive id count)

```
E.g. There are drive0 and drive1 available.

$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.14
SNMPv2-SMI::enterprises.45065.1.7.14.1.1.0 = STRING: "0"
SNMPv2-SMI::enterprises.45065.1.7.14.1.1.1 = STRING: "1"
```

Set free drive id
Must set RAID level first

```
E.g. RAID1 selected and must choose at least 2 drives

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.14.1.1.0 s "1"
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.14.1.1.1 s "1"
```

4. **Set createVolumeBasicData**
RAID level and free drive ids setup required

Controller Function createVolumeBasicData OID - 1.7.12.1.3.(controller id)

```
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.3.0 i 1
```

If **success**, there will be parameters **size**, **stripe size**, **access policy**, **disk cache policy** and **init state** available.
5. **Volume name**
:**Set volume name**

Name OID - 1.7.15.1.0

```
E.g. Set volume name "Test"

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.15.1.0 s "Test"
```

6. **Volume size**
:**Get available size**
Size OID - 1.7.16.1.0

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.16
SNMPv2-SMI::enterprises.45065.1.7.16.1.0 = INTEGER: 714880
```

Set volume size

```
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.16.1.0 i 714880
```

7. **Volume stripe size**
:**Get available stripe size**
Stripe Size OID - 1.7 17.1.1.(strip size options)

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.17
SNMPv2-SMI::enterprises.45065.1.7.17.1.1.0 = INTEGER: 64
SNMPv2-SMI::enterprises.45065.1.7.17.1.1.1 = INTEGER: 128
SNMPv2-SMI::enterprises.45065.1.7.17.1.1.2 = INTEGER: 256
SNMPv2-SMI::enterprises.45065.1.7.17.1.1.3 = INTEGER: 512
SNMPv2-SMI::enterprises.45065.1.7.17.1.1.4 = INTEGER: 1024
```

Set stripe size

```
E.g. Set stripe size 512.

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.17.1.1.3 i 1
```

8. **Volume access policy**
:**Get available access policy**
Access Policy OID - 1.7.18.1.1.(access policy options)

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.18
SNMPv2-SMI::enterprises.45065.1.7.18.1.1.0 = STRING: "Read Write"
SNMPv2-SMI::enterprises.45065.1.7.18.1.1.1 = STRING: "Read Only"
SNMPv2-SMI::enterprises.45065.1.7.18.1.1.2 = STRING: "Blocked"
```

Set volume access policy

```
E.g. Set access policy read write

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.18.1.1.0 s "1"
```

9. **Volume disk cache policy**
:**Get available disk cache policy**
Disk Cache Policy OID - 1.7.19.1.1.(disk cache policy options)

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.19
SNMPv2-SMI::enterprises.45065.1.7.19.1.1.0 = STRING: "Unchanged"
SNMPv2-SMI::enterprises.45065.1.7.19.1.1.1 = STRING: "Enable"
SNMPv2-SMI::enterprises.45065.1.7.19.1.1.2 = STRING: "Disable"
```

Set volume disk cache policy

```
E.g. Set disk cache policy enable

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.19.1.1.1 s "1"
```

10. **Volume init state**
:**Get available init state**
Init State OID - 1.7.20.1.1.(init state options)

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.20
SNMPv2-SMI::enterprises.45065.1.7.20.1.1.0 = STRING: "No Init"
SNMPv2-SMI::enterprises.45065.1.7.20.1.1.1 = STRING: "Quick Init"
SNMPv2-SMI::enterprises.45065.1.7.20.1.1.2 = STRING: "Full Init"
```

Set volume init state

```
E.g. Set init state quick init

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.20.1.1.1 s "1"
```

11. **Set createVolumeParam**
:Name, size, stripe size, access policy, disk cache policy and init state setup required
Controller Function createVolumeParam OID - 1.7.12.1.4.(controller id)

```
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.4.0 i 1
```

If **success**, there will be **new volume** available.

8.2.2.5 Delete volume

Set volume delete id
0: cancel, 1: set

Volume OID - 1.7.5.1.24.(volume id)

```
E.g. Set delete volume 1

snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.24.1 i 1
```

Trigger volume delete action
Must setup volume delete id first

Controller Function OID - 1.7.12.1.1.(controller id)

```
snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.12.1.1.0 i 1
```

8.2.2.6 Volume patch

1. **Patch options**
:**Name**
Volume Name OID - 1.7.5.1.2.(volume id)

```
E.g. Set name "Test1"

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.2.0 s "Test1"
```

Read policy
Read Policy OID - 1.7.5.1.15.(volume id)

1: read ahead, 2: no read ahead

```
E.g. set read policy no read ahead

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.15.0 s "2"
```

Write policy
Write Policy OID - 1.7.5.1.16.(volume id)

1: write through, 2: write back, 3: write cached

```
E.g. set write policy write through

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.16.0 s "1"
```

IO policy
IO Policy OID - 1.7.5.1.17.(volume id)

1: direct io, 2: cached io

```
E.g. set io policy direct io

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.17.0 s "1"
```

Access policy
Access Policy OID - 1.7.5.1.18.(volume id)

1: read write, 2: read only, 3: blocked

```
E.g. set access policy read write

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.18.0 s "1"
```

Disk cache policy
Disk Cache Policy OID - 1.7.5.1.19.(volume id)

1: enable, 2: disable

```
E.g. set disk cache policy enable

$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.19.0 s "1"
```

2. **Patch action**
Must have **at least one patch option** set first

Volume Patch Action OID - 1.7.5.1.25.(volume id)

0: cancel, 1: action

```
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address)
1.3.6.1.4.1.45065.1.7.5.1.25.0 i 1
```

If success, volume table information will be updated according to patch options setup.

8.2.2.7 RAID event log information

Get storageRaidEventController - storage RAID event controller(e.g. storage id,controller id: 1,1)

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.6.0
SNMPv2-SMI::enterprises.45065.1.7.6.0 = STRING: "1,1"
```

Get storageRaidEventCount - storage RAID event count

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.7.0
SNMPv2-SMI::enterprises.45065.1.7.7.0 = INTEGER: 15290
```

Get storageRaidEventStartIndex - storage RAID event start index

```
$ snmpget -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.8.0
SNMPv2-SMI::enterprises.45065.1.7.8.0 = INTEGER: 1
```

Get RAID event log list

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.9
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.1 = INTEGER: 1
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.2 = INTEGER: 2
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.3 = INTEGER: 3
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.33 = INTEGER: 33
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.34 = INTEGER: 34
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.35 = INTEGER: 35
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.1 = STRING: "Information"
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.2 = STRING: "Information"
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.3 = STRING: "Information"
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.33 = STRING: "Information"
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.34 = STRING: "Warning"
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.35 = STRING: "Information"
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.1 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:4
(h:m:s)"
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.2 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:4
(h:m:s)"
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.3 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:4
(h:m:s)"
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.33 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:51
(h:m:s)"
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.34 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:51
(h:m:s)"
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.35 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:51
(h:m:s)"
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.1 = STRING: "Test event: 'LogInit initial erase'"
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.2 = STRING: "Event log cleared"
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.3 = STRING: "Test event: 'Event log adjusted, possibly due Firmware version
incompatibility'"
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.33 = STRING: "Inserted: PD 07(e0x86/s6) Info: enclPd=86, scsiType=0,
portMap=0c, sasAddr=00000000013154d4,00000000592e6b79"
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.34 = STRING: "PD 07(e0x86/s6) is not supported"
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.35 = STRING: "Inserted: PD 08(e0x86/s10)"
```

Set storageRaidEventStartIndex - change event log start index

```
E.g. Change event log start index from 36 line
$ snmpset -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.8.0 i
36
```

Get event log list with changed start index

```
$ snmpwalk -v3 -l authPriv -u root -a SHA -A test123456 -x AES -X test123456 $(Server_IP_Address) 1.3.6.1.4.1.45065.1.7.9
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.1 = INTEGER: 36
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.1.35 = INTEGER: 70
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.1 = STRING: "Information"
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.2.35 = STRING: "Information"
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.1 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:0:51
(h:m:s)"
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.3.35 = STRING: " Time elapsed since power on : 0 years, 0 months, 0 days; 0:2:54
(h:m:s)"
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.1 = STRING: "Inserted: PD 08(e0x86/s10) Info: enclPd=86, scsiType=0,
portMap=05, sasAddr=00000000013154d4,00000000592f968d"
...
SNMPv2-SMI::enterprises.45065.1.7.9.1.4.35 = STRING: "Controller reset requested by host, completed"
```